On the robustness of the metric dimension to adding a single edge

A Preprint

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Abstract

The metric dimension (MD) of a graph is a combinatorial notion capturing the minimum number of landmark nodes needed to distinguish every pair of nodes in the graph based on graph distance. The MD has been connected with the number of sensor nodes required in several applications, including diffusion source detection, navigation and pattern matching. In this paper, we study how much the MD can change if we add a single edge to the graph. We find that the MD cannot decrease by more than two, however, there are examples where the increase can be very large. We believe that such a large increase can occur only in specially constructed graphs. For two simple families of graphs - ring graphs and grid graphs - we show that the increase of the MD cannot be larger than two either. For grid graphs, when the extra edge is sampled uniformly randomly, we almost completely characterize the limiting distribution of the MD.

1 Introduction

The metric dimension (MD) of a finite, simple graph is a combinatorial notion first defined in 1975 by [1] and independently by [9]. It can be interpreted as the minimum number of landmark nodes that can distinguish every pair of nodes based on the graph distances from these landmark nodes. The MD of d-dimensional grid graphs is d, hence for these graphs the MD is consistent with our common-sense notions of dimension. On the theoretical side, the MD deep connections to the automorphism group of the graph G [14, 15, 16], and hence the graph isomorphism problem [26]. In applications, the MD is used to compute the minimum number of landmark nodes required in robot navigation [12, 13], computational chemistry [2], and network discovery [5]. A recent application that is gaining more and more interest is the problem of finding patient zero of an epidemic. Finding patient zero can be especially useful in the early stages of an epidemic, as it was in the case of COVID-19 in the beginning of 2020 [28]. There are multiple mathematical models of the patient zero problem. The first model was introduced by [19], who were interested in finding the source of a rumour in a network. In this paper, we focus on the model of [20], who introduced the problem of detecting the first node of an epidemic given the underlying graph and the time of infection of small subset of sensor nodes. In the case of a deterministically spreading epidemic, the minimum number of sensors required to detect patient zero has been connected to the MD by [3]. In reality, epidemics are not deterministic, but the MD can still give good estimates about the number of sensors required in a noisy setting [25].

Since the MD is NP-hard to compute [12] and is approximable only to a factor of $\log(N)$ [5, 17], theoretical studies play an essential role in understanding the MD of large graphs. The MD of a wide range of combinatorial graph families have already been computed, we refer to [18] for a list of references. For applications on naturally forming networks like the patient zero detection problem, random graph models are the most appropriate tool for theoretical study. There are only a few results on the MD of random graphs, including Erdős-Rényi graphs [10] and a large class of random trees [21, 22]. In the case of $\mathcal{G}(n, p)$ Erdős-Rényi random graphs, it has been shown that the MD goes through a non-monotone, zig-zag behavior as we vary the probability of connections p, and we let the number of nodes n tend to infinity [10]. Not only is the behaviour non-monotone, it is also not smooth in the parameters. For example for $p = n^{-\frac{1}{2}}$ we have $\mathrm{MD} \approx \log(n)$ but for $p = n^{-\frac{1}{2}+\epsilon}$ we have $\mathrm{MD} \approx \sqrt{n}$. For $p = \Theta(1)$ we have $\mathrm{MD} \approx \log(n)$ again. This surprising result raises the main question of the current paper: how robust is the notion of the MD to the addition and deletion of edges? We ask this question both in general and for specific graphs (the ring and the 2D grid graph), with the edges added or deleted adversarially or randomly. To keep the analysis tractable, in this paper we only focus on single edge additions and deletions. We formulate our results in terms of edge additions and not removals, because in the case of arbitrary edge changes the two formulations are symmetric (non-robustness to edge removal can be seen as non-robustness to edge addition in the smaller graph), and in the case of random edge changes, adding edges is a more difficult problem to study than removing edges (we are usually interested in sparse graphs, and in sparse graphs there are more ways to add edges than to remove them).

Understanding the robustness of the MD to a singe edge addition or deletion has wide ranging practical implications. For the graphs in which the MD is non-robust, the MD might not be a very informative notion for application purposes. This is especially true in the application settings where we only have a noisy estimate of the underlying network. For instance, in most papers on patient zero detection, the contact network is assumed to be completely known; an assumption which does not hold in reality. Indeed, the contact network is usually estimated [32], which is a very challenging task [33]. With the exception of [35], we are not aware of any theoretical work in the source detection community that addresses the question of robustness in the estimation or the number of required sensors, when our knowledge of the contact network is noisy. We note that robustness to node failures has been more extensively studied, see [34] and the several follow-up articles.

In different applications, where we know the underlying network exactly, non-robustness of the MD can hint at opportunities for improvement or vulnerabilities to malicious attacks depending on our goal in the specific application. For instance, in the source obfuscation problem, our goal is to spread some information in a network so that a few spy nodes are not able to detect the information source [24, 23]. These source obfuscation models are used to anonymize transactions on the Bitcoin network [27]. Similarly, to the source detection problem, in source obfuscation the MD could serve as a proxy for how many spies are needed for the attacker to detect the source. Our result, that the MD can significantly increase when adding an edge and cannot increase too much when deleting an edge, suggests that we might be able to improve our privacy guarantees by edge additions. In a different scenario, when the attackers are able change the network topology (similarly to wormhole attacks [29]), a non-robust MD could help identify vulnerabilities. In this paper we find that such vulnerabilities can be substantial in the case of edge deletions, but are fairly benign in the case of edge additions.

Our proofs rely on careful combinatorial analysis, and a detailed description of how the shortest paths change in a graph after adding an edge. In particular, when adding edge to graph, we study the set of node pairs between which the shortest paths are changed and unchanged. These sets depend on the extra edge, but otherwise they are highly structured. We are not aware whether this structure (described in Section 2) has been previously studied in the literature, but we believe it could bring an insight into different problems where the addition of a single edge is studied (i.e. wormhole attacks [29] and the dynamic all pairs shortest paths problem in data structures [30, 31]).

1.1 Related work

Questions similar to ours were studied by [8], [35] and [2]. In [8], the authors define the notion of the *threshold dimension* of a graph G as the minimum MD we can achieve by adding an arbitrary number of edges to G. Obviously, adding too many edges will bring G close to the complete graph, which has a very large MD, but the authors show that for some graphs G it is possible to add a few edges in a smart way to significantly reduce the MD. The authors of [8] also connect the threshold dimension with the dimension of the Euclidean space in which the graph can be embedded.

In [35], we are given k connected graphs and it is assumed that $k-1$ edges are missing between them, which would connect all k components into a single one. The *extended metric dimension* is the number of landmarks we need to distinguish any pair of nodes, no matter where the $k-1$ edges are. Note that as opposed to our setup, in the setup of [35] the landmarks are placed non-adaptively to the extra edges, in fact, the nodes must be distinguished without knowing the location of the extra edges.

To the best of our knowledge, the question of how much the MD of a graph can change on the addition of a single edge has only been studied for trees. In [2], they found that on the addition of an arbitrary edge, the MD cannot increase by more than one and cannot decrease by more than two. We show that the decrease cannot be larger than two in *any* graph, however, in certain graphs, the increase can be very large.

1.2 Summary of results

Before summarizing the results we give the rigorous definition of the metric dimension.

Definition 1 (MD)**.** Let $G=(V,E)$ be a simple connected graph, and let us denote by $d_G(A,B)\in\mathbb{N}$ the length of the shortest path (that is, the number of edges) between nodes A and B that we call *graph distance*. A subset $R\subseteq V$ is a *resolving set* in G if for every pair of nodes $A\neq B\in V$ there is a distinguishing node $X\in R$ for which $d_G(A,X)\neq d_G(B,X)$. The minimal cardinality of a resolving set is the *metric dimension* of G, denoted by $\beta(G)$.

G	Adding an arbitrary edge e	Adding a random edge e
Arbitrary graph	$\beta(G)-2\leq\beta(G\cup\{e\})\leq\lvert V\rvert$	-
Ring	$\beta(G\cup\{e\})=2$	$\beta(G\cup\{e\})=2$
Grid ($n\times n$)	$2\leq\beta(G\cup\{e\})\leq 4$	$\lim_{n\to\infty}\mathbf{P}(\beta(G_n\cup\{e\})\geq 3)=1$ $\lim_{n\to\infty}\mathbf{P}(\beta(G_n\cup\{e\})=4)\leq\frac{8}{27}$

Figure 1: Overview of our results.

Our results are summarized in Figure 1. We show that for arbitrary graphs, the MD cannot decrease by more than two (Section 3.1), however, the increase can be very large (Section 3.2). In particular, we show an example of a graph where adding a particular edge increases the MD from $\Theta(\log(N))$ to $\Theta(N)$. We also give a heuristic argument to show that such anomalies can happen only if the graph is heterogeneous, in the sense that it has subgraphs where nodes cannot be easily distinguished by the nodes close-by, but they can be easily distinguished from distant nodes (Section 3.3).

In more homogeneous networks, such as the ring graph (Section 4) or the grid graph (Section 5), we show that the MD cannot increase too much even if we add an arbitrary edge, similarly to tree networks. The ring graph is an interesting example where adding one edge cannot change the MD. It is not surprising that the MD of the ring does not decrease since the MD of ring is 2, and it is well-known that the only graph resolved by a single node is the path. The reason for why the MD cannot increase either is that the ring graph has a large number of resolving sets of size two to begin with, and it is not possible to destroy all of them.

Finally, in the case of the square grid graph, we show that the MD can be two, three and four. We also show that if the extra edge is sampled uniformly randomly, then the MD is at least three with probability tending to one, and that the probability that the MD is four is bounded by $8/27$ (Theorem 3). We conjecture that $8/27$ *equals* the probability that the MD of the grid augmented by an edge is four, and thus the limiting distribution of the MD is $3+\mathrm{Ber}(8/27)$, where Ber is the Bernoulli distribution. The only part missing from proving this conjecture is a lower bound on the MD, when the extra edge is in a specific configuration. Such lower bound proofs are especially tedious, since one must show that no set of landmark nodes of a certain size can distinguish every pair of nodes, which often leads to a long case-by-case analysis. Instead, we proved as much as we could reasonably write down in a paper, and the rest of our results are stated as a conjecture at the end of the paper (Conjecture 1). A similar approach was used in [4] when determining the MD of torus graphs.

2 Changes in the all-pairs shortest paths after adding an edge

In this section we will develop tools to understand how the shortest paths change in a graph after adding an extra edge.

Let $G = (V, E_G)$ be a connected simple graph, with vertex set V (we use the word vertex and point interchangeably) and edge set E_G. We add an edge e between two non-adjacent vertices E and F to obtain a graph $G' = (V, E_G \cup \{e\})$. Let $d_H(A, B)$ denote the length of the shortest path between vertices A and B in graph H. For simplicity, we will use the notation $d_G(A, B) = AB$.

Remark 1. If we want to reach vertex B from vertex A, there are three options: Either we do not use e at all, or we use e from E to F or we use e from F to E. Hence,

$$d_{G'}(A, B) = \min(AB, AE + 1 + FB, AF + 1 + EB). \tag{1}$$

Clearly, we cannot increase the distance between two vertices by adding an edge, or in other words either $d_{G'}(A, B) \leq AB$. Next, we describe the pairs of vertices whose distance decreased after adding the edge.

Definition 2 (special region)**.** For any vertex A, $R_A = \{Z \in V \mid d_{G'}(Z, A) < ZA\}$. We will refer R_A as the *special region of* A.

The special region contains the vertices which will "use" the extra edge e to reach A. Formally, we can write this as $Z \in R_A$ is equivalent with $d_{G'}(A, Z) = \min(AE + 1 + FZ, AF + 1 + EZ) < ZA$.

Definition 3 (normal region, normal vertex)**.** $N_A = V \setminus R_A$ will be referred as the *normal region of* A. We call the intersection of all normal regions as simply the *normal region* and we denote it by N. A vertex in the normal region is called a *normal vertex*.

The normal region can be succinctly expressed as $N = \{Z \in V \mid R_Z = \varnothing\}$. For a normal vertex $Z \in N$ we have $d_{G'}(A, Z) = AZ$ for every vertex A, that is distances from or to these vertices Z are unchanged after adding edge e, which makes normal vertices the simplest type of vertices from the point of view of our analysis. The following claim helps us to characterize the normal region for any graph.

Claim 1. The set of vertices V can be partitioned to the following three sets,

$$\begin{aligned} R_E &= \{A \in V \mid AE - AF > 1\} \\ N &= \{A \in V \mid |AE - AF| \leq 1\} \\ R_F &= \{A \in V \mid AE - AF < -1\}. \end{aligned}$$

The intuition for Claim 1 is that if we are trying to reach A from some other node, we may want to use e in the EF direction if F is closer to A, we may want to use e in the FE direction if E is closer to A, and there is no gain in using e if E and F are almost equidistant to A. The three regions are illustrated in Figure 2.

Proof. First assume that $|AE - AF| \leq 1$. For an arbitrary vertex B in the graph, using triangular inequality,

$$AB \leq AE + EB \leq AF + 1 + EB.$$

Similarly,

$$AB \leq AE + 1 + FB.$$

Hence, by Remark 1, we have that $d_{G'}(A, B) = AB$. As this is true for any vertex $B \in V$, we must have $A \in N$.

Next assume $AE - AF > 1$. Then, by Remark 1,

$$d_{G'}(A, E) = \min(AE, AE + 1 + AF, AF + 1) = AF + 1,$$

which implies that $A \in R_E$. The $AE - AF < -1$ case follows analogously. $\square$

The usefulness of partitioning the vertices into R_E, N and R_F goes beyond just characterizing the normal region. Note that R_E collects the vertices that use the edge in the FE direction (because F is closer to them), and R_F collects the vertices that use the edge in the EF direction. There are no nodes that use the extra edge in both directions. Hence, if the two nodes are in the same special region R_E or R_F, they are using the extra edge in the same direction, and they cannot use the extra edge to reduce the distance between themselves. We formalize this intuition in the next claim.

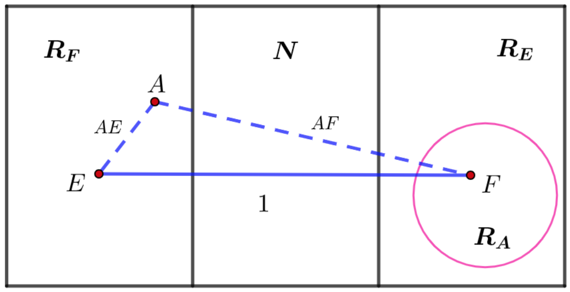


Figure 2: Graph G' partitioned into three regions: R_E, N and R_F.

Claim 2. If two vertices A and B lie in the same special region R_E or R_F, then $d_{G'}(A,B) = d_G(A,B)$, or equivalently $B \notin R_A$ and $A \notin R_B$.

Proof. Without loss of generality, let $A, B \in R_E$. Then, we have $AE - AF > 1$ and $BE - BF > 1$ by Claim 1. Therefore,

$$d_{G'}(A,B) = \min(AB, AE + 1 + FB, AF + 1 + EB) = AB,$$

because

$$AE + 1 + FB > AF + 2 + FB \geq AB + 2,$$

and

$$AF + 1 + EB > AF + 2 + FB \geq AB + 2$$

by the triangle inequality. □

Remark 2. Containment in special regions defines an anti-reflexive, symmetric and anti-transitive (never transitive) relation between pairs of vertices. Containment in normal regions defines a reflexive, symmetric and intransitive (not necessarily transitive) relation between pairs of vertices.

Proof. For both special and normal regions (anti-)reflexivity follows from the definition and symmetry follows from the symmetry of distances in both G and G'. The anti-transitivity of special regions follows from Claim 2. Indeed, if $A \in R_B$ and $B \in R_C$, then the pairs (A,B) and (B,C) are in different special regions R_E or R_F, which implies that A and C must be both in R_E or R_F and we cannot have $A \in R_C$. □

We are now ready to justify the illustration in Figure 2.

Remark 3. For a vertex $A \in R_F$ we have

$$F \in R_A \subseteq R_E,$$

and similarly, for a vertex $B \in R_E$ we have

$$E \in R_B \subseteq R_F.$$

Proof. For a vertex $A \in R_F$, the statement $F \in R_A$ follows by the symmetric nature of special regions (Remark 2). The $R_A \subseteq R_E$ is a simple consequence of anti-transitivity. Indeed, $R_A \cap N$ is empty by definition, and $R_A \cap R_F$ is empty because we cannot have $Z \in R_F$, $Z \in R_A$ and $A \in R_F$ all hold at the same time. □

Next, we use the anti-transitivity property to make equation (1) more explicit.

Claim 3. For any $A, B \in V$, we have

$$d_{G'}(A,B) = \begin{cases} AE + 1 + FB & \text{if } A \in R_B \text{ and } A \in R_F \\ AF + 1 + EB & \text{if } A \in R_B \text{ and } A \in R_E \\ AB & \text{otherwise, i.e., } A \in V \setminus R_B = N_B. \end{cases} \tag{2}$$

Proof. We consider only the case $A \in R_B$ and $A \in R_F$; the second case is symmetric, and the third holds by definition. By the definition of special regions, $A \in R_F$ is equivalent with

$$d_{G'}(A, F) = \min(AF + 1 + EF, AE + 1 + FF) < AF,$$

which further implies $AE + 1 < AF$.

By the anti-transitivity of special regions, $A \in R_B$ and $A \in R_F$ together imply $B \in R_E$, which is equivalent with

$$d_{G'}(B, E) = \min(BE + 1 + FE, BF + 1 + EE) < BE,$$

which further implies $BF + 1 < BE$.

Finally, $A \in R_B$ is equivalent with

$$d_{G'}(A, B) = \min(AE + 1 + FB, AF + 1 + EB),$$

which reduces to $d_{G'}(A, B) = AE + 1 + FB$ since $AE < AF$ and $BF < BE$. □

We already used the intuition that vertices in special regions "gain" from the addition of the extra edge. We formalize this intuition in the next definition.

Definition 4 (Gain, $\mathrm{Gain}_{\max}$)**.** Let the decrease in the distance between two vertices due to edge e be denoted as

$$\mathrm{Gain}(A, B) = AB - d_{G'}(A, B).$$

Let the maximum gain associated to a node A be denoted as

$$\mathrm{Gain}_{\max}(A) = \max_X(\mathrm{Gain}(A, X)).$$

Remark 4. For vertex $A \in R_E$, vertex E gets the maximum benefit of the extra edge to reach A, that is, $\mathrm{Gain}_{\max}(A) = \mathrm{Gain}(A, E) = AE - (1 + AF)$. More generally, for any $A \in V$, we have

$$\mathrm{Gain}_{\max}(A) = \max(0, |AF - AE| - 1).$$

We can also observe that, by Claim 1, $A \in N$ if and only if $\mathrm{Gain}_{\max}(A) = 0$. A similar statement hold for vertex F instead of E.

Proof. Suppose for contradiction that there is a node $B \in V$ for which $\mathrm{Gain}(A, B) > \mathrm{Gain}(A, E)$. Since $A \in R_E$, this node B must be in R_F, otherwise by Claim 2 we have $\mathrm{Gain}(A, B) = 0$. Then, the following inequalities must hold:

$$\begin{aligned}
\mathrm{Gain}(A, B) &> \mathrm{Gain}(A, E)\\
AB - d_{G'}(A, B) &> AE - d_{G'}(A, E)\\
AB - (BE + 1 + AF) &> AE - (1 + AF)\\
AB &> AE + BE.
\end{aligned}$$

The last inequality above contradicts the triangle inequality, and the proof is completed. □

3 General graphs

The focus of this paper is to understand, how much the metric dimension can change on adding a single edge. We will show that the metric dimension of the new graph cannot decrease by more than two (Section 3.1), however the increase can be exponentially large (Section 3.2).

The surprisingly large increase in the metric dimension is shown by adding a specific edge to a carefully constructed graph. Our main idea is to construct a graph, in which the vertices of R_F can be efficiently distinguished only by some vertices in R_E (but not by vertices in R_F). Then, after adding edge e, the vertices in R_E can reach R_F on new shortest paths, and they will not distinguish vertices in R_F anymore. Hence R_F will have to be distinguished by vertices in R_F, which will require significantly more nodes.

We conclude the section by showing that our example for a large increase is universal in the sense that we prove a tight upper bound on $\beta(G')$ using (roughly speaking) the metric dimension of R_E and R_F (Section 3.3).

3.1 Lower bound on the decrease of the metric dimension

Lemma 1. Let $G = (V, E)$ be a connected graph, and let G' be the graph obtained by adding edge e between vertices E and F as before. Let S be the resolving set of G'. Then, $S \cup \{E, F\}$ is the resolving set for G, and consequently $\beta(G) \leq \beta(G') + 2$.

Proof. Consider any pair of vertices X and Y. Let $Z \in S$ be a vertex that distinguishes this pair in G'. There are three possibilities for where X and Y can be located: they can be both N_Z, both in R_Z, or (without loss of generality) $X \in R_Z$ and $Y \in N_Z$. Observe that if Z is a normal vertex, then both X and Y must belong to N_Z, hence only Case 1 below is possible. If, however, Z is not a normal vertex, then without loss of generality we can assume $Z \in R_E$. Now we treat the three cases for the location of X and Y.

Case 1: X and Y both belong to N_Z.

In this case, $d_{G'}(X, Z) = d_G(X, Z)$ and $d_{G'}(Y, Z) = d_G(Y, Z)$ so Z will distinguish the pair in G too.

Case 2: X and Y both belong to R_Z.

Since we assumed $Z \in R_E$, by the anti-transitivity of special regions (Remark 2) $X, Y \in R_F$. Then, by Claim 3,

$$d_{G'}(X, Z) = XE + 1 + FZ,$$

and similarly,

$$d_{G'}(Y, Z) = YE + 1 + FZ.$$

Since $d_{G'}(X, Z) \neq d_{G'}(Y, Z)$, we must also have $XE \neq YE$. Hence, E will distinguish X and Y in G.

Case 3: X is in R_Z and Y is in N_Z.

Since we assumed $Z \in R_E$, we have that $X \in R_F$ by the anti-transitivity of special regions. By the definition of special and normal regions, we have

$$XZ > d_{G'}(X, Z) \overset{(2)}{=} XE + 1 + FZ \tag{3}$$

$$YZ = d_{G'}(Y, Z) \overset{(1)}{\leq} YE + 1 + FZ. \tag{4}$$

Assume that none of Z and E distinguishes X and Y in G. Then, $XE = YE$ and $XZ = YZ$, which clearly gives a contradiction between inequalities (3) and (4). Hence, one of Z and E must distinguish X and Y.

$\square$

3.2 An example with an exponential increase in the metric dimension

In the following example, we give a construction for a graph on $3n + \lceil \log_2(n) \rceil - 1$ nodes, in which the increase in the metric dimension is at least $n - \lceil \log_2(n) \rceil - 3$ on adding a single (specific) edge. The construction is shown in Figure 3 for $n = 8$.

Graph $G^\star$ has 6 levels indexed by $l \in \{-1, \dots, 4\}$. Levels 1-3 each contain $n - 1$ vertices, which are indexed by i for each level. Level 0 contains $\lceil \log_2(n) \rceil$ vertices indexed by j. Levels -1 and 4 contain the single vertices $F = v_1^{(-1)}$ and $E = v_1^{(4)}$. We connect all of the vertices of level 0 and 3 to F and E, respectively. We connect the vertices of level 1 (respectively, level 2) to the vertices of level 2 (resp., level 3) if and only if the vertices of both levels 1-2 (resp., 2-3) have the same index. Finally, we connect a vertex labelled i in level 1 to a vertex labelled j in level 0 if and only if the j^{th} bit in the binary representation of i is one. For example, $v_1^{(1)}$ is connected only to $v_{\lceil \log_2(n) \rceil}^{(0)}$ because binary representation of 1 is $0 \dots 01$. This construction leads therefore to the following definition.

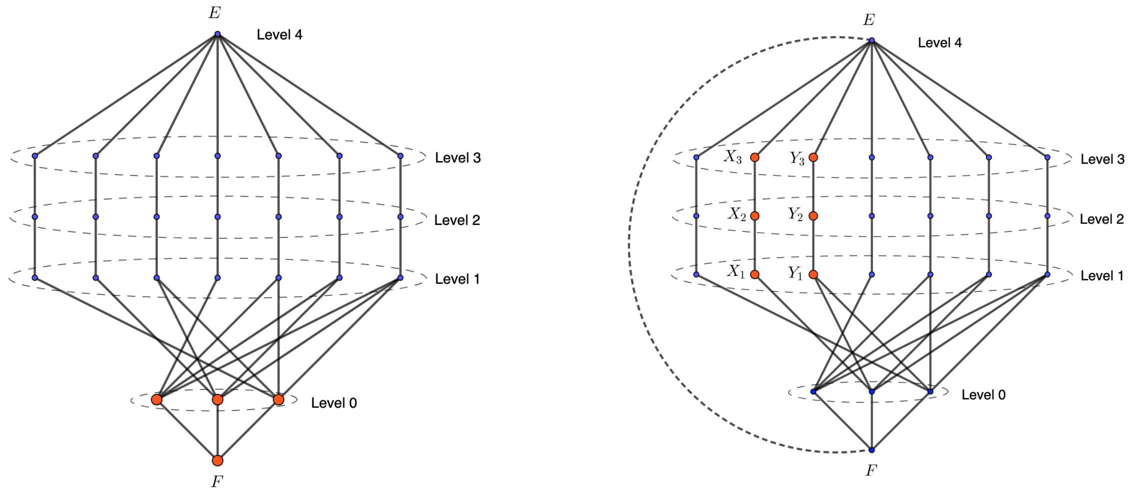
Definition 5 ($G^\star$)**.** For $n > 1$, let $G^\star = (V^\star, E_{G^\star})$, with

$$V^\star = \left\{ v_j^{(0)} \mid j \in \{1, \dots, \lceil \log_2(n) \rceil\} \right\} \cup \left\{ v_i^{(l)} \mid l \in \{1, 2, 3\}, i \in \{1, \dots, n-1\} \right\} \cup \{E, F\},$$

$$E_{G^\star} = \left\{ (F, v_j^{(0)}) \right\} \cup \left\{ (v_j^{(0)}, v_i^{(1)}) \mid \mathrm{bin}(i)_j = 1 \right\} \cup \left\{ (v_i^{(l)}, v_i^{(l+1)}) \mid l \in \{1, 2\} \right\} \cup \left\{ (v_i^{(3)}, E) \right\},$$

where $\mathrm{bin}(i)_j$ denotes the j^{th} bit of the binary representation of the number i.

Claim 4. The set $S^\star = \left\{ v_j^{(0)} \right\} \cup F$ resolves $G^\star$. Consequently, $\beta(G^\star) \leq \lceil \log_2(n) \rceil + 1$.



(a) G

(b) G'

Figure 3: Example where MD increases by a large amount (for $n = 8$)

Proof. We need to show that any pair of vertices in $V^\star \setminus S^\star$ are distinguished. There are two possibilities for any pair of distinct vertices: either they are in different levels or in the same level. If they are on different levels, vertex F will distinguish them, because for any $v_i^{(l)}$ with $l \in \{1, 2, 3, 4\}$ we have $d_{G^\star}(v_i^{(l)}, F) = l + 1$. If they are on the same level, the binary representations of their index i will differ at at least one position. Let the j^{th} bit of both labels be different. Then, vertex $v_j^{(0)}$ will distinguish them, because its distance to the vertex whose label has the j^{th} bit equal to 1 is two hops shorter than its distance to the vertex whose label has the j^{th} bit equal to 0. Therefore all pairs of points are distinguished, which completes the proof. ☐

Now we add an edge e between vertices E and F. The resulting graph $G^{\star\prime}$ is shown in Figure 3b.

Claim 5. The metric dimension of graph $G^{\star\prime}$ is at least $n - 2$.

Proof. Notice that the set of nodes that can distinguish $v_j^{(3)}$ and $v_k^{(3)}$ is

$$\left\{ v_i^{(l)} \mid l \in \{1, 2, 3\}, i \in \{j, k\} \right\}.$$

This is because all other nodes can reach both $v_j^{(3)}$ and $v_k^{(3)}$ through E on their shortest path and E cannot distinguish any pair of nodes on level 3. Hence, distinguishing nodes on level 3 is equivalent to resolving a star graph, and the metric dimension of $G^{\star\prime}$ is at least $n - 2$. ☐

Combining Claims 4 and 5, we observe that the increase in the metric dimension of $G^\star$ on adding e is at least $n - \lceil \log_2(n) \rceil - 3$.

3.3 Upper bound on the increase of the metric dimension

While in the previous section we saw that $\beta(G')$ can be exponentially larger than $\beta(G)$, we can provide an upper bound on $\beta(G')$ in terms of the MD of the subgraphs of G.

Lemma 2. Let $G = (V, E)$ be a connected graph, and let G' be the graph obtained by adding edge e between vertices E and F as before. Let $V_1 = \{U \in V \mid d_G(U, E) \leq d_G(U, F)\}$ and $V_2 = \{U \in V \mid d_G(U, E) \geq d_G(U, F)\}$. Let G_1 and G_2 be subgraphs of G induced on vertex sets V_1 and V_2, respectively. Then,

$$\beta(G') \leq \beta(G_1) + \beta(G_2) + 2.$$

Proof. Let S_1 and S_2 be the resolving sets of minimum size of graphs G_1 and G_2, respectively. We prove that $S = S_1 \cup S_2 \cup \{E, F\}$ is a resolving set of G'. Let $N_1 = \{U \in V \mid 0 \leq d_G(U, F) - d_G(U, E) \leq 1\}$ and

$N_2 = \{U \in V \mid 0 \leq d_G(U, E) - d_G(U, F) \leq 1\}$. By Claim 1, $N_1, N_2 \subseteq N$, where N is the normal region. Consider two vertices X and Y. There are two cases:

Case 1: $X, Y \in V_1$ or $X, Y \in V_2$.

Without loss of generality, let $X, Y \in V_1$. Let $A \in S_1$ be the vertex which resolves X and Y in G_1. Since $V_1 = R_F \cup N_1$ and $V_2 = R_E \cup N_2$, by Claim 2, $d_G(X, A) = d_{G'}(X, A)$ and $d_G(Y, A) = d_{G'}(Y, A)$, hence X and Y are resolved by A in G', too.

Case 2: $X \in V_1 \setminus V_2$ and $Y \in V_2 \setminus V_1$ or vice and versa.

Without loss of generality, let $X \in V_1 \setminus V_2$ and $Y \in V_2 \setminus V_1$. Note that by definition, $V_1 \setminus V_2$ and $V_2 \setminus V_1$ contain the nodes that are closer to E and F, respectively. Hence, we can always go through e when going from $V_1 \setminus V_2$ to F or $V_2 \setminus V_1$ to E on a shortest path, that is

$$d_{G'}(X, F) = 1 + d_{G'}(X, E) \tag{5}$$
$$d_{G'}(Y, E) = 1 + d_{G'}(Y, F). \tag{6}$$

Assume for contradiction that none of E and F distinguish X and Y. This implies that $d_{G'}(X, E) = d_{G'}(Y, E)$ and $d_{G'}(X, F) = d_{G'}(Y, F)$. Adding both equations gives

$$d_{G'}(Y, E) + d_{G'}(X, F) = d_{G'}(Y, F) + d_{G'}(X, E).$$

Substituting values from (5) and (6) gives a contradiction. Hence, either E or F will distinguish these two vertices.

For every possible pair of vertices we showed a distinguishing vertex in S. Finally,

$$\beta(G') \leq |S| = |S_1| + |S_2| + 2 = \beta(G_1) + \beta(G_2) + 2.$$

□

Next we present a graph $G^{\star\star}$ for which the upper bound of Lemma 2 is achieved. The graph has 74 vertices and it is drawn on Figure 4. The four solid black nodes labelled as E in the figure represent a single vertex E in the graph $G^{\star\star}$. Similarly, the four solid black nodes labelled F represent vertex F. All other nodes shown in the figure represent distinct nodes. The graph $G^{\star\star\prime}$ is obtained by adding an edge between vertices E and F. In this setting, $G_1^{\star\star}$, defined in Lemma 2, will be the sub-graph induced by nodes having green and blue outlines and $G_2^{\star\star}$ will be that induced by nodes having yellow and orange outlines.

Claim 6. $\beta(G_1^{\star\star}) = \beta(G_2^{\star\star}) = 8$ and $\beta(G^{\star\star\prime}) = 18$.

Proof. Notice that $G_1^{\star\star}$ and $G_2^{\star\star}$ are isomorphic, hence their metric dimensions must be equal as well. First we show $\beta(G_1^{\star\star}) \leq 8$. Indeed we have 8 triangles in $G_1^{\star\star}$, and selecting one degree 2 vertex in each triangle is enough to distinguish any two vertices. To show $\beta(G_1^{\star\star}) \geq 8$, observe that we need to select one vertex from each of the triangles. The equality $\beta(G_1^{\star\star}) = \beta(G_2^{\star\star}) = 8$ together with Lemma 2 proves $\beta(G^{\star\star\prime}) \leq 18$.

Next, we show that $\beta(G^{\star\star\prime}) \geq 18$. Again, notice that $G^{\star\star\prime}$ contains 16 triangles, and we must select a vertex in each of them. Notice that even after we selected these 16 nodes, the solid colored pairs in Figure 4 are not distinguished. Moreover, it is not possible to distinguish all 4 of these solid colored pairs by adding a single vertex to the set. Indeed, if any of the blue stroked nodes are selected, the blue solid pair is not distinguished. A similar argument holds for all other colors. This shows that we must add at least two nodes to the initial 16, and the metric dimension of $G^{\star\star\prime}$ is at least 18, which completes the proof. □

4 Ring graph

It is well-known that the metric dimension of a ring graph on n nodes is two. A set of any two adjacent vertices is a resolving set. In this section we will see that metric dimension will remain two on adding an edge between any two vertices. Hence, this is an example of a graph whose metric dimension cannot be changed by adding an edge.

Theorem 1. Let G be a ring with n vertices. G' is obtained by adding an edge between any two vertices of G. Then, the metric dimension of G' is exactly two.

Proof. Let us denote the vertices of G by $A_1, \dots, A_n$. Without loss of generality, let one endpoint of the extra edge be A_1 and the other be A_x with $1 < x < n$. It is well-known (see e.g. [2]) that the metric dimension of any graph that is not a path is at least two. We show that set of vertices $X = A_{\lfloor \frac{x}{2} \rfloor}$ and $Y = A_{\lfloor \frac{x}{2} \rfloor + 1}$ is a resolving set for G' which proves that metric dimension is exactly two.

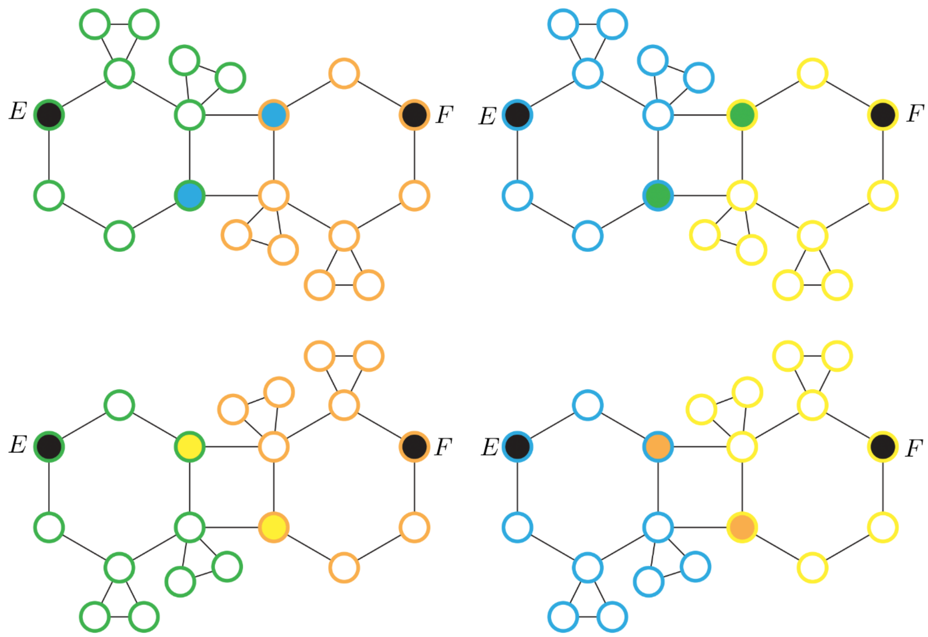


Figure 4: Graph $G^{\star\star}$ and points E, F for which upper bound is achieved

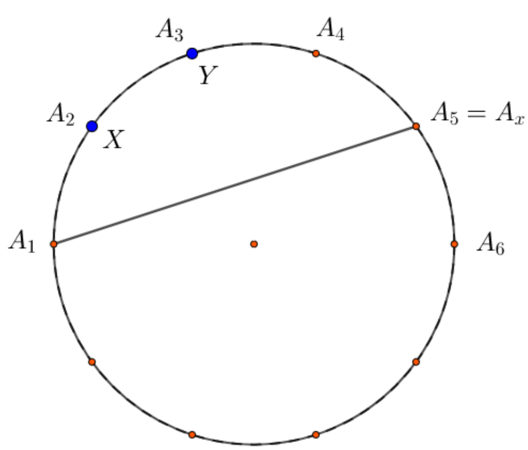


Figure 5: Ring with an additional edge. Vertices X and Y marked as blue points form a resolving set.

We distinguish two cases based on the parity of x.

Case 1: x is even:

In this case,

$$\begin{aligned}|d_G(A_1, X) - d_G(A_x, X)| &= \left|\min\left(\frac{x}{2} - 1, n - \frac{x}{2} + 1\right) - \min\left(x - \frac{x}{2}, n - x + \frac{x}{2}\right)\right| \\ &= \left|\left(\frac{x}{2} - 1\right) - \left(x - \frac{x}{2}\right)\right| = 1,\end{aligned}$$

and

$$|d_G(A_1,Y)-d_G(A_x,Y)| = \left|\min\left(\frac{x}{2}, n-\frac{x}{2}\right) - \min\left(x-\frac{x}{2}-1, n-x+\frac{x}{2}+1\right)\right|$$
$$= \left|\left(\frac{x}{2}\right) - \left(x-\frac{x}{2}-1\right)\right| = 1.$$

By Claim 1, both X and Y are normal vertices. Consequently, the extra edge has no effect on distances of any vertex from X and Y. Since a set of two adjacent vertices is a resolving set for a ring, the set $\{X, Y\}$ is a resolving set for G' too.

Case 2: x is odd:

In this case,

$$|d_G(A_1,X)-d_G(A_x,X)| = \left|\min\left(\frac{x-1}{2}-1, n-\frac{x-1}{2}+1\right) - \min\left(x-\frac{x-1}{2}, n-x+\frac{x-1}{2}\right)\right|$$
$$= \left|\left(\frac{x-1}{2}-1\right) - \left(x-\frac{x-1}{2}\right)\right| = 2,$$

and

$$|d_G(A_1,Y)-d_G(A_x,Y)| = \left|\min\left(\frac{x-1}{2}, n-\frac{x-1}{2}\right) - \min\left(x-\frac{x-1}{2}-1, n-x+\frac{x-1}{2}+1\right)\right|$$
$$= \left|\left(\frac{x-1}{2}\right) - \left(x-\frac{x-1}{2}-1\right)\right| = 0.$$

By Claim 1, Y is a normal vertex and X is not. By Remark 4,

$$\text{Gain}_{\max}(X) = \max(0, |d_G(A_1,X)-d_G(A_x,X)|-1) = 1, \tag{7}$$

or equivalently $0 \le \text{Gain}(A,X) \le 1$ for any vertex A in the ring.

Suppose for contradiction that there exist two nodes M and N that are distinguished by neither X nor Y in G', i.e., $d_{G'}(M,X) = d_{G'}(N,X)$ and $d_{G'}(M,Y) = d_{G'}(N,Y)$. Then, since Y is a normal vertex, we must have $d_G(M,Y) = d_G(N,Y)$, and since we also know that $\{X,Y\}$ is a resolving set in G, we must have $d_G(M,X) \neq d_G(N,X)$. Let $d_G(M,X) < d_G(N,X)$ without loss of generality, and let us denote $k = d_G(M,Y) = d_G(N,Y)$. Next, we compute the distances between X and M, N. Since M and N are equidistant from Y in G, and X is the neighbor of Y that is closer to M, X must on the shortest path between Y and M in G, and therefore we must have $d_G(M,X) = k-1$. Now we have two cases: either Y is on the shortest path between X and N, in which case we must have $d_G(N,X) = k+1$, or Y is not on the shortest path between X and N, which is only possible if X and Y are the two farthest vertices from N, which further implies that $d_G(N,X) = d_G(N,Y) = k = (n-1)/2$, with n odd. We treat these sub-cases separately.

Sub-case 1: $d_G(N,X) = k$

This sub-case is only possible for n odd and $k = (n-1)/2$, which means that that M and N are the two farthest vertices from Y, and are therefore adjacent. Also A_1 must be on the YM half, and A_x must be on the YN half of the ring. Since the pairs of vertices (M,N) and (A_1,A_x) are both equidistant from Y, the distance between M and A_1 and the distance between N and A_x will also be equal in G. Formally, we write

$$d_G(M,A_1) = d_G(M,Y) - d_G(A_1,Y) = d_G(N,Y) - d_G(A_x,Y) = d_G(N,A_x).$$

Since A_1 is on the YM half, and A_x is on the YN half of the ring, the shortest path between M and A_x must go through N, and the shortest path between N and A_1 must go through M. We also know that M and N are adjacent, and therefore

$$|d_G(M,A_1) - d_G(M,A_x)| = |d_G(M,A_1) - (d_G(N,A_x)+1)| = 1, \tag{8}$$

and

$$|d_G(N,A_1) - d_G(N,A_x)| = |(d_G(M,A_1)+1) - d_G(N,A_x)| = 1. \tag{9}$$

Equations (8) and (9) show, by Claim 1, that M and N are normal vertices. Hence,

$$|d_{G'}(M,X) - d_{G'}(N,X)| = |d_G(M,X) - d_G(N,X)| = |(k-1)-k| = 1,$$

contradicting the assumption $d_{G'}(M,X) = d_{G'}(N,X)$.

Sub-case 2: $d_G(N, X) = k + 1$

Since $\text{Gain}_{\max}(X) = 1$ by equation (7), we have:

$$0 \leq d_G(M, X) - d_{G'}(M, X) \leq 1, \tag{10}$$

and

$$0 \leq d_G(N, X) - d_{G'}(N, X) \leq 1. \tag{11}$$

Inequalities (10) and (11), with the assumption of $d_{G'}(M, X) = d_{G'}(N, X)$, imply that

$$-1 \leq d_G(M, X) - d_G(N, X) \leq 1,$$

which is impossible because previously we had $d_G(N, X) = k + 1$ and $d_G(M, X) = k - 1$.

Hence, all pairs of nodes are distinguished by X and Y, and the proof is completed. □

5 Grid graph

The main technical result of this paper is on the metric dimension of the grid graph augmented with one edge.

Let $G = (V, E_G)$ be a rectangle grid graph with m rows and n columns. Let the tuple (i, j) denote the vertex in i^{th} column and j^{th} row. The upper left, upper right, bottom right, bottom left corners are labeled as

$$P = (1, 1), \qquad Q = (n, 1), \qquad R = (n, m), \qquad S = (m, 1),$$

respectively (see Figure 7). For grid G and vertices $A = (x_A, y_A)$ and $B = (x_B, y_B)$, we denote the distance

$$d_G(A, B) = AB = |x_A - x_B| + |y_A - y_B|.$$

Let e be the edge between vertices $E = (x_E, y_E)$ and $F = (x_F, y_F)$ with $x_E, x_F \in \{1, ..., n\}$, $y_E, y_F \in \{1, ..., m\}$, with the assumption that $EF \geq 2$. Let $G' = (V, E_G \cup \{e\})$ be the grid augmented with one edge.

Assumption 1 (symmetry breaking)**.** We assume that:

1. $x_F \leq x_E$
2. $y_E \leq y_F$
3. $x_E - x_F \leq y_F - y_E$.

Geometrically, this means that the edge is tilted right, F is below and to the left of E, and the angle between the edge and the horizontal axis is between 45 and 90 degrees (see Figure 7). If the edge is in any other orientation, we can flip or rotate the grid horizontally and/or vertically to bring the edge in this orientation, hence Assumption 1 can be made without loss of generality.

5.1 Deterministic upper bound

Theorem 2. Let $G = (V, E)$ be a rectangle grid graph with m rows and n columns. For an edge e between any two nodes in V, let $G' = (V, E \cup \{e\})$. Then, the set of all 4 corners of the original grid is a resolving set for G', and consequently $\beta(G') \leq 4$.

Proof. We start by making observations about which special regions the four corners P, Q, R, S belong to. First, notice that

$$QF - QE = (n - x_F) + (y_F - 1) - (n - x_E) - (y_E - 1) = EF \geq 2,$$

Hence by Claim 1, $Q \in R_F$. Similarly, $S \in R_E$.

Then, notice that

$$PF - PE = (x_F - 1) + (y_F - 1) - (x_E - 1) - (y_E - 1) = (y_F - y_E) - (x_E - x_F) \geq 0$$

where the last inequality holds by Assumption 1. Claim 1 implies therefore that P belongs to either R_F or N. Similarly, R belongs to either R_E or N. In any case, by Claim 2, it can be deduced that

$$(R_P \cup R_Q) \cap (R_S \cup R_R) = \varnothing \tag{12}$$

In fact, it turns out that $R_P \cup R_Q = R_E$ and $R_R \cup R_S = R_F$, but we are not showing this because it is not needed in this proof. Instead, let $R_W = V \setminus \{R_P \cup R_Q \cup R_R \cup R_S\}$ (the white region in Figure 7), and we note that the sets $R_P \cup R_Q$, $R_S \cup R_R$ and R_W partition the set of nodes V.

To prove the theorem, for any pair of nodes A, B, we are going to assign two of the corners $\{P, Q, R, S\}$ in the resolving set, and we are going to show that one of the two must distinguish A and B. The assignment will depend on whether A and B belong to $R_S \cup R_R$, $R_P \cup R_Q$ or R_W. Moreover, we further divide the region $R_S \cup R_R$ to $R_R \setminus R_S$, $R_R \cap R_S$ and $R_S \setminus R_S$, and the region $R_P \cup R_Q$ to $R_Q \setminus R_P$, $R_Q \cap R_P$ and $R_P \setminus R_Q$, and we treat each subregion separately.

This would mean treating $7 \cdot 7 = 49$ cases, but we make some simplifications. Let us suppose that the first point A is in R_W or in R_Q. The cases when A falls in R_P, R_R or R_S are very similar. We make no assumptions on where B falls, but combine similar cases. Finally, we arrive to 8 cases, which are presented in Figure 6. The table shows the various possibilities of regions where A and B can belong to (denoted by R_1 and R_2), the corresponding pair of corners which distinguish A and B, and the claim which proves this.

R_1	R_2	Distinguishing Corners	Claim used
R_W	$R_W \cup R_P \cup R_Q$	R, S	8
R_W	$R_R \cup R_S$	P, Q	8
$R_Q \setminus R_P$	$R_W \cup R_Q \cup R_P$	R, S	8
$R_Q \setminus R_P$	R_S	Q, S	7
$R_Q \setminus R_P$	$R_R \setminus R_S$	Q, R	8
$R_Q \cap R_P$	$R_W \cup R_Q \cup R_P$	R, S	8
$R_Q \cap R_P$	R_S	Q, S	7
$R_Q \cap R_P$	R_R	P, R	7

Figure 6: The assignment of corners to the pair A, B, when $A \in R_1$ and $B \in R_2$.

We conclude the proof by stating and proving Claims 7 and 8. $\square$

Claim 7. If $A \in R_Q$ and $B \in R_S$ then $d_{G'}(A, Q) \neq d_{G'}(B, Q)$ or $d_{G'}(A, S) \neq d_{G'}(B, S)$, i.e., A and B are distinguished by the opposite corners Q and S. Similarly, if $A \in R_P$ and $B \in R_R$, then they are distinguished by P and R.

Proof. Suppose for contradiction that $d_{G'}(A, Q) = d_{G'}(B, Q)$ and $d_{G'}(A, S) = d_{G'}(B, S)$.

Since $A \in R_Q$, $A \notin R_S$, $B \in R_S$ and $B \notin R_Q$, we have

$$BQ = d_{G'}(B, Q) = d_{G'}(A, Q) = AF + 1 + EQ$$
$$AS = d_{G'}(A, S) = d_{G'}(B, S) = BE + 1 + FS.$$

Adding these equations gives

$$BQ + AS = AF + EQ + BE + FS + 2. \tag{13}$$

Applying the triangle inequality to points B, E, Q and A, F, S and adding both the inequalities, we get

$$BQ + AS \leq BE + EQ + AF + FS,$$

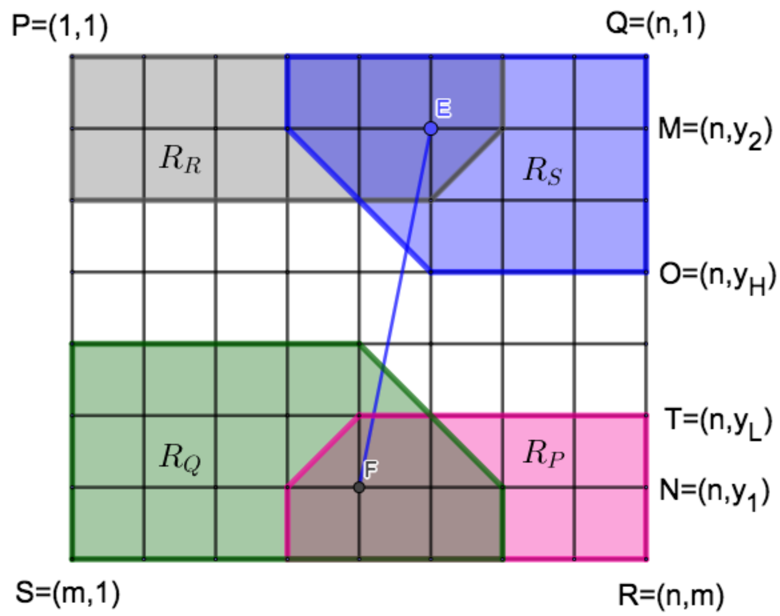


Figure 7: The sets R_P, R_Q, R_R, R_S, R_W are colored grey, blue, pink, green and white, respectively. Vertices on the boundary of coloured regions are included in the respective coloured region.

which contradicts (13). A similar proof holds for $A \in R_P$ and $B \in R_R$ with corners P and R. $\square$

Claim 8. If two vertices A, B are outside of the union of the special regions of two adjacent corners, then they are distinguished by those two corners. For example, if $A, B \in V \setminus \{R_P \cup R_Q\}$ then P and Q distinguish A and B.

Proof. The distances from A, B to P, Q in G' are same as that in G, and we know [7] that the set of two adjacent corners is a resolving set of a rectangle grid. $\square$

5.2 Adding a random edge

Theorem 2 tells us that the metric dimension of a grid and one extra edge must take a value from the set $\{2, 3, 4\}$, and in fact, all three values can occur. In Conjecture 1, we present a set of conditions, which we believe completely characterize the metric dimension of a grid and one extra edge, but proving this conjecture seems tedious. Instead, we are interested in a probabilistic approach: what is the distribution of the metric dimension when a uniformly randomly selected edge is added?

First we define some quantities which will be useful for the remaining section.

Definition 6 (Gain′)**.** Let

$$\mathrm{Gain} = \mathrm{Gain}(E, F) = |y_F - y_E| + |x_E - x_F| - 1$$

as in Definition 4, and let

$$\mathrm{Gain}' = \max(0, ||y_F - y_E| - |x_E - x_F|| - 1) = \mathrm{Gain}(F, (x_F, y_E)).$$

The notion of Gain captures the maximum gain for any pair of nodes. The pair of vertices (E, F) obviously have maximum gain, however, there can be other pairs which have the same gain. For two vertices X, Y, let us denote by $\mathrm{Rec}(X, Y)$ the rectangle that has opposite corners X and Y, and sides parallel to the sides of the grid. Then, the pairs (A, B), with $A \in \mathrm{Rec}(E, Q)$, and $B \in \mathrm{Rec}(F, S)$ also have $\mathrm{Gain}(A, B) = \mathrm{Gain}$, since there is a shortest path between A and B in G that passes through both E and F. The notion of Gain' has a very similar interpretation as Gain. We defined Gain' as the gain between vertices F and (x_F, y_E). Notice that (x_F, y_E) is also a corner of $\mathrm{Rec}(E, F)$. Therefore, while Gain is about the gain between the opposite corners, Gain' is about the gain between the adjacent corners of the same rectangle (by symmetry the gain between E and (x_E, y_F) is also Gain'). Similarly to Gain, there are many other pairs of vertex pairs (A, B) with $\mathrm{Gain}(A, B) = \mathrm{Gain}'$. These are the pairs (A, B) with $A \in \mathrm{Rec}(P, (x_F, y_E))$ and $B \in \mathrm{Rec}(F, R)$, and symmetrically the pairs with $A \in \mathrm{Rec}(R, (x_E, y_F))$ and $B \in \mathrm{Rec}(E, P)$. Roughly speaking, we could thus say that Gain is useful if we want to measure the distance between vertex pairs with one vertex close to S and the other close to Q, while Gain' is useful if we want to measure the distance between vertex pairs with one vertex close to P and the other close to R.

One of the key steps of the main proof in this section will be about treating the case when Gain' is very small. This is the case when the extra edge has (or is close to having) a 45 degree angle with the sides of the grid, and $\mathrm{Rec}(E, F)$ is (or is close to being) a square. In the extreme case, when $\mathrm{Gain}' = 0$, no vertex pairs close to P and R use the extra edge, and the structure of the special and normal regions are different from the case when $\mathrm{Gain}' \geq 1$. When $\mathrm{Gain}' = 1$, there are still some subtle but inconvenient structural differences compared to the $\mathrm{Gain}' \geq 2$ case. Fortunately, since we are adopting a probabilistic framework, in the proof we will be able to ignore the cases with $\mathrm{Gain}' \leq 1$, as these cases have a vanishing probability of occurring.

Definition 7. Let $\mathbf{P}_n$ be the probability distribution over potential extra edges $e_n = ((x_E, y_E), (x_F, y_F))$ that we can add to G_n, where (x_E, y_E) and (x_F, y_F) are two uniformly random vertices of G_n.

Theorem 3. Let G_n be the $n \times n$ grid and let $G'_n = G_n \cup \{e_n\}$ with e_n sampled from distribution $\mathbf{P}_n$. Then, the following results hold:

$$\lim_{n\to\infty} \mathbf{P}_n(\beta(G'_n) \in \{3, 4\}) = 1 \tag{14}$$

$$\lim_{n\to\infty} \mathbf{P}_n\left(\beta(G'_n) = 3 \;\middle|\; \mathrm{Gain}' \text{ is odd or } \min(|x_E - x_F|, |y_E - y_F|) < \frac{\mathrm{Gain}'}{2} + 2\right) = 1 \tag{15}$$

$$\lim_{n\to\infty} \mathbf{P}_n\left(\mathrm{Gain}' \text{ is odd or } \min(|x_E - x_F|, |y_E - y_F|) < \frac{\mathrm{Gain}'}{2} + 2\right) = \frac{19}{27}. \tag{16}$$

According to Theorem 3, the asymptotic probability that the MD of the square grid with an extra edge is three is at least $19/27$. We believe that it is also true that the MD is at least four when Gain' is even and $x_E - x_F \geq \mathrm{Gain}'/2 + 2$. If we could prove this, we could state that the asymptotic probability of $\beta(G')$ being three is *exactly* $19/27$, and $\beta(G') \to \mathrm{Ber}(8/27) + 3$ in probability, where $\mathrm{Ber}(q)$ is a Bernoulli random variable with parameter q. We believe that a brute-force approach similar to the proof of Theorem 2 can work, but it requires a tedious case-by-case analysis that is out of scope of this paper.

The probabilistic formulation of Theorem 3 allows us to ignore the edge-cases that would be too tedious to check individually, but it introduces new challenges as well. In rest of this section, we explore these new challenges and we reduce equations (14)-(16) to technical Lemmas 3, 4 and 5, which are of deterministic nature. We give the proof of Theorem 3 at the end of this section, but we defer the proof of the technical lemmas to Section 5.4.

The specific edge-cases that we ignore using the probabilistic formulation are given in Assumption 2.

Assumption 2 (edge-case removal)**.** We assume that

1. $x_F \neq x_E$
2. $\mathrm{Gain}' \geq 2$
3. none of E and F lie on the boundary of the grid.

In addition to Assumption 2, we are also going to make use of Assumption 1 as we did in the proof of Theorem 2. Assumptions 1 and 2 applied together have some additional implications.

Remark 5. Assumption 1 and 2 together imply that

1. $x_F < x_E$
2. $y_E < y_F$.
3. $x_E - x_F < y_F - y_E$

Using Assumption 1 in the probabilistic formulation is not as straightforward anymore, as symmetry breaking can also break the uniformity of the sampling of the extra edge. Indeed, sampling a random edge that satisfies Assumption 1 is not the same as sampling an edge from $\mathbf{P}_n$ and rotating and reflecting it so that Assumption 1 is satisfied. In Claims 9 and 11, we are going to show that after removing only $O(n^3)$ edges from $V \times V$, and thus slightly changing the distribution $\mathbf{P}_n$, the symmetry breaking will not violate the uniformity of the sampling anymore.

Definition 8 ($\mathcal{P}$,$\mathcal{Q}$,$\tilde{\mathbf{P}}_n$, $\mathbf{Q}_n$)**.** Let $\mathcal{P}$ be the set of extra edges $((x_E, y_E), (x_F, y_F))$ that satisfy Assumption 2, and let $\mathcal{Q}$ the set of extra edges that satisfy both Assumptions 1 and 2. Let $\tilde{\mathbf{P}}_n$ and $\mathbf{Q}_n$ be the uniform probability distribution over $\mathcal{P}$ and $\mathcal{Q}$, respectively.

In Claim 9 we show that $\mathbf{P}_n$ is close to $\tilde{\mathbf{P}}_n$, and in Claim 11 we show that $\tilde{\mathbf{P}}_n$ is close to $\mathbf{Q}_n$. These two claims allow us to use $\mathbf{Q}_n$ instead of $\mathbf{P}_n$ in the proof of Theorem 3.

Claim 9. For $\mathbf{P}_n$ and $\tilde{\mathbf{P}}_n$ given in Definitions 7 and 8,

$$\lim_{n\to\infty} \|\mathbf{P}_n - \tilde{\mathbf{P}}_n\|_{TV} = 0.$$

Proof of Claim 9. The support of $\mathbf{P}_n$ is $V \times V$, and $|V \times V| = n^4$ because each of the four coordinates x_E, y_E, x_F and y_F can take four values. Recall, that $\mathcal{P} \subset (V \times V)$, and the set $(V \times V) \setminus \mathcal{P}$ consists of the edges that do not satisfy Assumption 2. Therefore, to upper bound the cardinality of $(V \times V) \setminus \mathcal{P}$, it is enough to upper bound the number of edges violating each of the conditions in Assumption 2. It is clear that the number of edges that violate the first condition is n^3; the coordinates x_E, y_E, y_F can be chose arbitrarily n^3 different ways, and then setting $x_F = x_E$ gives exactly one unique edge that violates the first condition. For a more insightful but less precise explanation, notice that the original set $V \times V$ had four degrees of freedom, and we lost one to violating the condition, hence we are left with three degrees of freedom and $O(n^3)$ edges. It is not hard to see that we lose one degree of freedom to violate the second and third conditions as well, and therefore the number of edges violating these conditions are also $O(n^3)$. We conclude that the number of edges in $(V \times V) \setminus \mathcal{P}$ are also of order $O(n^3)$.

Then,

$$\begin{aligned}
2\|\mathbf{P}_n - \tilde{\mathbf{P}}_n\|_{TV} &= \sum_{e\in\mathcal{P}} |\mathbf{P}_n(e) - \tilde{\mathbf{P}}_n(e)| + \sum_{e\in V\times V\setminus\mathcal{P}} \mathbf{P}_n(e) \\
&= |\mathcal{P}| \left| \frac{1}{|V \times V|} - \frac{1}{|\mathcal{P}|} \right| + \frac{|(V \times V) \setminus \mathcal{P}|}{|V \times V|} \\
&= (n^4 + O(n^3)) \left| \frac{1}{n^4} - \frac{1}{n^4 + O(n^3)} \right| + \frac{O(n^3)}{n^4} \\
&= O\left(\frac{1}{n}\right).
\end{aligned}$$

$\square$

Definition 9 (H)**.** Let us consider the following actions on the extra edges of the grid:

1. by h_1 the reflection along the vertical line through the midpoints of sides PQ and SR,
2. by h_2 the reflection along the horizontal line through the midpoints of sides PS and QR,
3. and by h_3 switching the two endpoints of the edge.

Let H be the group generated by h_1, h_2 and h_3 acting on the edges.

Notice that group H acting on the edges is isomorphic to the $\mathbb{Z}_2^3$ group. Indeed, all three actions have order two and commute with each other. Thus, H can be described as $\{h_1^i h_2^j h_3^k \mid i, j, k \in \{0, 1\}\}$. Also, notice that for $e \in \mathcal{Q}$, applying h_1, h_2 and h_3 flips the inequality labelled with the same index in Remark 5, and keeps the other two inequalities unchanged.

Definition 10. Let h be a map, which for each edge $e \in \mathcal{Q}$ returns the set of edges that we get by applying the elements of H to e.

The sets $h(e)$ can be seen as orbits of the edges under the action of H.

Claim 10. With $\mathcal{P}$, $\mathcal{Q}$ and h given in Definitions 8 and 10, the following three statements must hold:

1. $|h(e)| = 8$ for every $e \in \mathcal{Q}$
2. the orbits of the edges in $\mathcal{Q}$ are disjoint, i.e., $h(e_1) \cap h(e_2) = \varnothing$ for $e_1 \neq e_2 \in \mathcal{Q}$
3. for every $e \in \mathcal{P}$, there is an $e_2 \in \mathcal{Q}$ with $e \in h(e_2)$.

Proof of Claim 10. Statement 1 follows from the observation that every non-trivial group action in H flips a different subset of the inequalities in Remark 5, and two edges cannot coincide if they satisfy different sets of inequalities. For statement 2, since H is a group, if two orbits $h(e_1), h(e_2)$ have a non-empty intersection, we must have $e_1 \in h(e_2)$. However, every non-trivial group action in H flips at least one of the inequalities of Remark 5, which implies that if we apply a non-trivial group action, the image of $e_2 \in \mathcal{Q}$ cannot be in $\mathcal{Q}$. For statement 3, for edge $e \in \mathcal{P}$, let $\mathbf{v}(e) \in \{0,1\}^3$ be a binary vector, whose i^{th} entry indicates that e violates inequality i in Remark 5. Then $h_1^{\mathbf{v}(e)_1} h_2^{\mathbf{v}(e)_2} h_3^{\mathbf{v}(e)_3}$ is a group action that flips exactly the inequalities that are violated by e, and thus maps e into $\mathcal{Q}$. Let the image of e under this action be e_2, and then indeed, $e \in h(e_2)$. $\square$

Claim 11. Let $\mathcal{A}_n$ be a sequence of events defined on graph G_n' that are closed under the action of H. Then,

$$\lim_{n\to\infty} |\mathbf{P}_n(\mathcal{A}_n) - \mathbf{Q}_n(\mathcal{A}_n)| = 0.$$

Proof of Claim 11. The three statements of Claim 10 together imply that the orbits $h(e)$ of $e \in \mathcal{Q}$ partition $\mathcal{P}$ into sets of cardinality 8. A simple corollary is that $|\mathcal{P}| = 8|\mathcal{Q}|$.

Let us suppose that event $\mathcal{A}_n$ is closed under the action of H, or formally as $e \in \mathcal{A}_n$ implies $h(e) \subset \mathcal{A}_n$. This closedness property, combined with Claim 10 implies that the edges in $\mathcal{A}_n$ can also be counted as 8 times the number of edges in $\mathcal{A}_n \cap \mathcal{Q}$. Then,

$$\tilde{\mathbf{P}}_n(\mathcal{A}_n) = \frac{|\mathcal{A}_n|}{|\mathcal{P}|} = \frac{8|\mathcal{A}_n \cap \mathcal{Q}|}{8|\mathcal{Q}|} = \mathbf{Q}_n(\mathcal{A}_n). \tag{17}$$

Finally, we combine equation (17) with Claim 9 as

$$\lim_{n\to\infty} |\mathbf{P}_n(\mathcal{A}_n) - \mathbf{Q}_n(\mathcal{A}_n)| = \lim_{n\to\infty} |\mathbf{P}_n(\mathcal{A}_n) - \tilde{\mathbf{P}}_n(\mathcal{A}_n)| \leq \lim_{n\to\infty} \|\mathbf{P}_n(\mathcal{A}_n) - \tilde{\mathbf{P}}_n(\mathcal{A}_n)\|_{TV} = 0,$$

and the proof is completed. $\square$

Now we have all the ingredients to prove Theorem 3.

Proof of Theorem 3. Since all events in the statement of Theorem 3 are closed under the action of H on the square grid, Remark 11 shows that it is enough to prove equations (14)-(16) for distribution $\mathbf{Q}_n$. Note that because of statements 1 and 3 of Remark 5, $\min(|x_E - x_F|, |y_E - y_F|) = x_E - x_F$ for edges in $\mathcal{Q}$. Hence, the second condition in (15) and (16) reduces to $x_E - x_F < \text{Gain}'/2 + 2$ for distribution $\mathbf{Q}_n$.

The rest of the proof relies on Lemmas 3-5 given in Section 5.4, which have purely deterministic nature. Lemma 3 shows that for extra edges in $\mathcal{Q}$ (that is edges satisfying Assumption 1 and 2), the metric dimension of G' will be at least three deterministically, which, combined with Theorem 2, gives equation (14). Lemma 4 shows that there exists a resolving set of cardinality three for every extra edge in $\mathcal{Q}$ with an odd Gain'. For the extra edges in $\mathcal{Q}$ with an even Gain' and with $x_E - x_F < \text{Gain}'/2 + 2$, there exist different resolving sets of cardinality three, which is proved in Lemma 5. Thus, Lemmas 3, 4 and 5 combined imply equation (15).

Finally, we show equation (16). Let us denote by C the subset of vertex pairs in $\mathcal{Q}$ that satisfy the condition in equation (16), i.e.,

$$C = \left\{ (E,F) \in \mathcal{Q} \;\middle|\; \text{Gain}' \text{ is odd or } x_E - x_F < \frac{\text{Gain}'}{2} + 2 \right\}.$$

Let the complement of C be

$$\bar{C} = (V \times V) \setminus C = \left\{ (E,F) \in \mathcal{Q} \;\middle|\; \text{Gain}' \text{ is even and } x_E - x_F \geq \frac{\text{Gain}'}{2} + 2 \right\}.$$

Next, we calculate $|\bar{C}|/|\mathcal{Q}|$. Let $x_E - x_F = a$ and $y_F - y_E = b$, which together with Assumption 2 gives

$$b - a - 1 = \text{Gain}'.$$

Then, the conditions on a, b that need to be satisfied for an edge to be in $\bar{C}$ can be reformulated as :

1. $b - a$ is odd (equivalent to Gain' is even)
2. $b - a \geq 3$ (equivalent to $\text{Gain}' \geq 2$)
3. $a \geq \frac{b}{3} + 1$ (equivalent to $x_E - x_F \geq \frac{\text{Gain}'}{2} + 2$)

4. $1 \leq a, b \leq n-2$, as the extra edge is not horizontal nor vertical, and does not touch the boundary of the grid.

Let $b - a = 2i + 1$ with $i \geq 1$. With this parameterization, the first two conditions are already obviously satisfied. Substituting $a = b - 2i - 1$ into $a \geq b/3 + 1$ gives $b \geq 3(i + 1)$. Hence, for a fixed i, b can have values from $3(i + 1)$ to $n - 2$, and consequently, the maximum value that i can take is $\lfloor (n - 5)/2 \rfloor$. Note that for a given pair (a,b), there are $(n - a - 1)(n - b - 1)$ possible edges in G_n which do not touch the boundary. Therefore,

$$|\bar{C}| = \sum_{i=1}^{\lfloor \frac{n-5}{3} \rfloor} \sum_{b=3(i+1)}^{n-2} (n - b + 2i)(n - b - 1),$$

which reduces asymptotically to

$$|\bar{C}| = \frac{1}{27}n^4 + O(n^3).$$

Therefore,

$$\mathbf{Q}_n(\bar{C}) = \frac{|\bar{C}|}{|\mathcal{Q}|} = \frac{\frac{1}{27}n^4 + O(n^3)}{\frac{1}{8}n^4 + O(n^3)} = \frac{8}{27} + O\left(\frac{1}{n}\right), \tag{18}$$

Hence, $\mathbf{Q}_n(C) = 1 - \mathbf{Q}_n(\bar{C}) \to 19/27$, which shows equation (16) and completes proof of the theorem. □

In the rest of this section we state and prove Lemmas 3-5, which we will do in Section 5.4. Before introducing these lemmas, we prove some claims that will be useful later. We start by simple claims in this subsection, then in Section 5.3 we prove more involved results that charaterize the normal and special regions of G'.

The following claim shows that resolving sets must have nodes on the boundaries of the grid, which helps us reduce the number of subsets that we must prove are non-resolving.

Claim 12. If R is any resolving set of G' (the grid with extra edge EF) satisfying Assumption 1 and 2, there must be two vertices X and Y in R which satisfy following two properties:

1. They are on opposite boundaries of G'
2. If one of them is a corner, the other one must be an adjacent corner.

Proof. Consider vertices $A = (1, 2)$ and $B = (2, 1)$. It is easy to see that only vertices on boundaries PQ and PS except corner P will be able to distinguish A and B as none of E and F is on the boundaries. So we need at least one vertex on the union of the boundaries PQ and PS, excluding P, in the resolving set. A similar argument holds for the other 4 corners, hence we can deduce the two required conditions. □

Claim 13. For all $A, B \in V$ if $\mathrm{Gain}(A, B)$ is positive, it will have same parity as Gain and Gain' as defined in Definition 6.

Proof. Note that $\mathrm{Gain}(A, B) > 0$ indicates that A uses e to reach B. For this to happen, we must have $A \in R_F$ and $B \in R_E$ (or the other way around), in which case $d_{G'}(A, B) = AE + 1 + FB$. This gives

$$\begin{aligned}\mathrm{Gain}(A, B) &= AB - (AE + 1 + FB) \\ &= |x_A - x_B| + |y_A - y_B| - (|x_A - x_E| + |y_A - y_E| + 1 + |x_F - x_B| + |y_F - y_B|),\end{aligned}$$

which has same parity as

$$(x_A - x_B) + (y_A - y_B) - ((x_A - x_E) + (y_A - y_E) + 1 + (x_F - x_B) + (y_F - y_B)) = (y_E - y_F) - (x_E - x_F) - 1,$$

which has the same parity as $\mathrm{Gain} = (y_F - y_E) + (x_E - x_F) - 1$ and $\mathrm{Gain}' = (y_F - y_E) - (x_E - x_F) - 1$. □

Remark 6. Consider a vertex X and its 4 neighbouring vertices X_1, X_2, X_3, X_4. A single vertex in the graph cannot distinguish all of these 4 vertices.

Proof. Suppose that vertex A distinguishes all 4 vertices. By triangular inequality, AX_i can only take 3 distinct values, namely $AX - 1$, AX and $AX + 1$. Hence, by pigeon hole principle, at least 2 vertices will have same distance to A. □

5.3 Exact characterization of normal and special regions

The following quantities will be useful to describe the shape of the normal region.

Definition 11 (α, β)**.** Let

$$\alpha = \frac{1 + y_F + y_E + x_F - x_E}{2}$$

and

$$\beta = \frac{1 + y_F + y_E + x_E - x_F}{2}.$$

Remark 7. We make the following observations about α and β under Assumptions 1 and 2.

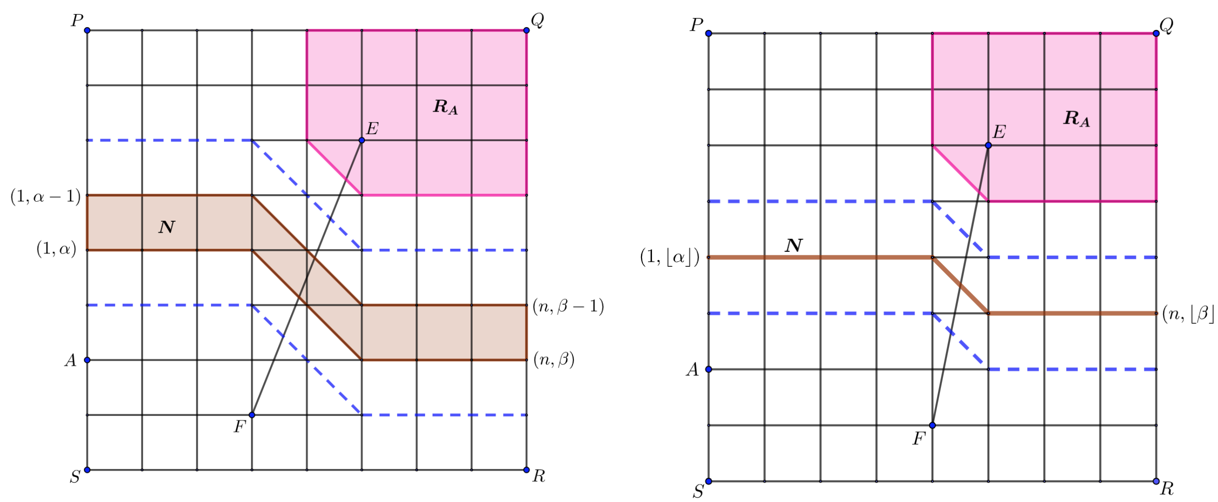
1. Note that $\beta - \alpha = x_E - x_F$. Assumptions 1 and 2 imply $x_E > x_F$, and since x_E, x_F are both integers, we know that $\beta - \alpha \geq 1$. Consequently, $\lfloor \beta \rfloor > \lfloor \alpha \rfloor$ holds.

2. Using Assumptions 1 and 2, we find the following useful equalities and inequalities:

$$\alpha = y_F - \frac{\text{Gain}}{2} = y_E + \frac{\text{Gain}' + 2}{2} \geq y_E + 2, \tag{19}$$

and

$$\beta = y_E + \frac{\text{Gain} + 2}{2} = y_F - \frac{\text{Gain}'}{2} \leq y_F + 1. \tag{20}$$



(a) When Gain' is even, the height of N is 2.

(b) When Gain' is odd, the height of N is 1.

Figure 8: Illustration for Claims 14 and 15. Brown region(including the boundary), which is just a set of line segments in the case when Gain' is odd, is the normal region of the grid. Pink region(including the boundary) indicates the special region of a point A belonging to R_E which lies on boundary PS of G'.

Next, we express precisely the normal region of the grid.

Claim 14. Under Assumptions 1 and 2,

$$\begin{aligned} N = &\{(x, y) \mid x < x_F, \alpha - 1 \leq y \leq \alpha\} \cup \\ &\{(x, y) \mid x_F \leq x \leq x_E, x_F - \alpha \leq x - y \leq x_F - \alpha + 1\} \cup \\ &\{(x, y) \mid x > x_E, \beta - 1 \leq y \leq \beta\} \end{aligned} \tag{21}$$

Geometrically, the normal region will be the union of three strips of "height" 1 or 2: two horizontal strips with y-coordinates around α and β respectively, and a third strip at 45 degree angle joining the two horizontal strips (see Figure 8). By "height" here we mean the number of vertices corresponding to each x coordinate. The height of the strips depends on whether α and β are integers or not (and thus on the parity of Gain'): when Gain' is even, α and β are integers and the height of the strip is 2; when Gain' is odd, α and β are odd integers divided by two, and the height of the strip is 1.

The union of the three strips forms a single continuous strip, which separates the grid into two connected components along the y-axis. The y-coordinates of the strip lie completely between the y-coordinates of nodes E and F, which means that for every x-coordinate, the normal vertices are sandwiched between non-normal vertices along the y-axis. Here we rely heavily on the inequality $\text{Gain}' \geq 2$ in Assumption 2; for $\text{Gain}' = 0$ the normal region can touch the PQ and RS boundaries of the grid.

Proof of Claim 14. Let $A = (x_A, y_A)$ be a vertex in N. By Claim 1, being a normal vertex is equivalent to

$$|AE - AF| = ||x_A - x_F| + |y_A - y_F| - |x_A - x_E| - |y_A - y_E|| \leq 1. \tag{22}$$

First we show that we cannot have $y_A < y_E$. If $y_A < y_E$, equation (22) reduces to

$$-1 \leq (y_F - y_E) + |x_A - x_F| - |x_A - x_E| \leq 1. \tag{23}$$

Next, by the triangular inequality we have

$$-(x_E - x_F) \leq |x_A - x_F| - |x_A - x_E| \leq (x_E - x_F),$$

and therefore by Definition 6,

$$\text{Gain}' + 1 \leq (y_F - y_E) + |x_A - x_F| - |x_A - x_E| \tag{24}$$

Due to the assumption $\text{Gain}' \geq 2$, equations (23) and (24) contradict each other. Hence, we cannot have $y_A < y_E$. Similarly it can be shown that we cannot have $y_A > y_F$. In short, for A to be a normal point, we must have $y_E \leq y_A \leq y_F$ (i.e., the y_A coordinate must be between E and F). This reduces equation (22) to

$$-1 \leq |x_A - x_F| + y_F - y_A - |x_A - x_E| - y_A + y_E \leq 1 \tag{25}$$

Now we are going to have three cases depending on whether $x_A < x_F$, $x_A > x_E$ or $x_F \leq x_A \leq x_E$. When $x_A < x_F < x_E$, equation (25) reduces to

$$\alpha - 1 = \frac{1 + y_F + y_E + x_F - x_E}{2} - 1 \leq y_A \leq \frac{1 + y_F + y_E + x_F - x_E}{2} = \alpha,$$

where α is given in Definition 11. This gives the first line of equation (21). Similarly, it can be verified that for the other two possibilities $x_F \leq x_A \leq x_E$ and $x_E < x_A$, we get the remaining two lines. □

Now that N is explicitly written in terms of the coordinates of the nodes, we can leverage the partitioning in Claim 1 to do the same for R_E and R_F. However, we find it more instructive to express R_E and R_F implicitly using N, instead of explicit equations similar to equation (21).

Remark 8. Under Assumptions 1 and 2,

$$R_F = \{(x, y) \notin N \mid \exists k \in \mathbb{N} \text{ with } (x, y + k) \in N\}, \tag{26}$$

and

$$R_E = \{(x, y) \notin N \mid \exists k \in \mathbb{N} \text{ with } (x, y - k) \in N\}. \tag{27}$$

Proof. By Claim 14, the normal region splits V into two connected components, one containing E, which we denote by V_F, and one containing F, which we denote by V_E. Now we show that $V_E = R_E$ and $V_F = R_F$. By Claim 1 it is clear that we cannot have two neighboring vertices A, B with $A \in R_E$ and $B \in R_F$. Indeed the equations $AE - AF > 1$, $BE - BF < -1$, $|AE - BE| \leq 1$ and $|AF - BF| \leq 1$ cannot hold at the same time. By Claim 1, the vertices $V \setminus N$ are partitioned into R_E and R_F, and since we cannot have two neighboring vertices split between R_E and R_F, each connected component V_E and V_F must be contained entirely in R_E or R_F. We also know that $E \in R_F$ and $F \in R_E$, which implies that the only way to assign the vertices of $V \setminus N$ into R_E and R_F is to have $R_E = V_E$ and $R_F = V_F$. □

In the next claim, we characterize the special regions of the nodes on the boundary PS of the grid. This will be useful in the subsequent results as we will be mainly dealing with nodes on the boundaries.

Claim 15. Let $A = (1, k)$ be a point on boundary PS, and $\text{Gain}_{\max}(A)$ given in Remark 4. Then, under Assumptions 1 and 2,

1. if A belongs to R_E, i.e., $k > \alpha$, with α given in Definition 11,

$$\begin{aligned} R_A = & \{(x,y) \mid x_E \le x, y \le y_E\} \cup \\ & \left\{(x,y) \,\middle|\, x_E \le x, 0 \le y - y_E < \frac{\mathrm{Gain}_{\max}(A)}{2}\right\} \cup \\ & \left\{(x,y) \,\middle|\, y \le y_E, 0 \le x_E - x < \frac{\mathrm{Gain}_{\max}(A)}{2}\right\} \cup \\ & \left\{(x,y) \,\middle|\, x \le x_E, y_E \le y, (x_E - x) + (y - y_E) < \frac{\mathrm{Gain}_{\max}(A)}{2}\right\}, \end{aligned} \tag{28}$$

and

$$\mathrm{Gain}_{\max}(A) = \begin{cases} \mathrm{Gain} & \text{for } k \ge y_F \\ \mathrm{Gain} - 2(y_F - k) & \text{for } \alpha < k < y_F \end{cases} \tag{29}$$

2. if A belongs to R_F i.e. $k < \alpha - 1$ and $\mathrm{Gain}' \ge 2$,

$$\begin{aligned} R_A = & \{(x,y) \mid x_F \le x, y_F \le y\} \cup \\ & \left\{(x,y) \,\middle|\, x_F \le x, 0 \le y_F - y < \frac{\mathrm{Gain}_{\max}(A)}{2}\right\} \cup \\ & \left\{(x,y) \,\middle|\, y_F \le y, 0 \le x_F - x < \frac{\mathrm{Gain}_{\max}(A)}{2}\right\} \cup \\ & \left\{(x,y) \,\middle|\, x \le x_F, y \le y_F, (x_F - x) + (y_F - y) < \frac{\mathrm{Gain}_{\max}(A)}{2}\right\}, \end{aligned} \tag{30}$$

and

$$\mathrm{Gain}_{\max}(A) = \begin{cases} \mathrm{Gain}' & \text{for } k \le y_E \\ \mathrm{Gain}' - 2(k - y_E) & \text{for } y_E < k < \alpha - 1. \end{cases} \tag{31}$$

By symmetry, the vertices $A = (n, k)$ on boundary QR have a similar expression for their special region, however, we do not include this in the paper in the interest of space.

We will only cover the $A \in R_F$ case; the other case is analogous. We are interested in the nodes $T \in R_A$, i.e., nodes that use edge e to reach A. By Remark 4, vertex E gets the maximum benefit from the extra edge, hence we expect R_A to be a neighbourhood "centered" at E. However, R_A cannot be a ball centered at E, because the directions are not equivalent. For instance, if T is in the rectangle formed by E and Q, then we can go to node E for "free", without sacrificing any of the gain we get by using the extra edge. This is because the shortest path from T to A in G passed through E anyways, so $\mathrm{Gain}(A,T) = \mathrm{Gain}_{\max}(A)$ (i.e., T also gets maximum benefit). Hence, all nodes in this rectangle will be in R_A. For a different example, if T is in the rectangle formed by E and P, then going along the y axis towards E is "free", but going along the x axis towards is a "detour", hence there may be a threshold for x_T below which the shortest path does not use the extra edge. We will make this intuition rigorous below.

Proof. First, we check that equation (29) agrees with the definition of $\mathrm{Gain}_{\max}$. By Remark 4, we have

$$\mathrm{Gain}_{\max}(A) = AE - (1 + AF), \tag{32}$$

which for $k \ge y_F$ implies

$$\mathrm{Gain}_{\max}(A) = (x_E - 1) + (k - y_E) - (1 + (x_F - 1) + (k - y_F)) = y_F - y_E + x_E - x_F - 1 = \mathrm{Gain}$$

because of Definition 4, and for $\alpha < k < y_F$ implies

$$\mathrm{Gain}_{\max}(A) = (x_E - 1) + (k - y_E) - (1 + (x_F - 1) + (y_F - k)) = \mathrm{Gain} - 2(y_F - k).$$

Next, we need to find nodes T such that

$$TE + 1 + FA < AT. \tag{33}$$

Combining equations (32) and (33) we get that

$$\begin{aligned} TE + AE - \mathrm{Gain}_{\max}(A) &< AT \\ TE_x + TE_y + AE_x + AE_y - \mathrm{Gain}_{\max}(A) &< AT_x + AT_y, \end{aligned} \tag{34}$$

where TE_x and TE_y denote the distance along the x and y axes, respectively (e.g., $TE_x = |x_T - x_E|$).

There are five cases depending on where T could be:

Case 1: T is in the rectangle formed by nodes E and Q

In this case there exists a shortest path in grid from T to A which passes through E, and T will certainly use the edge e to reach A. Hence, this rectangle belongs to R_A, which accounts for the first line of in equation (28).

Case 2: T is in the rectangle formed by E and P

In this case $AE_x = AT_x + TE_x$ and $AE_y = AT_y - TE_y$, which reduces equation (34) to

$$TE_x < \frac{\text{Gain}_{\max}(A)}{2}.$$

This accounts for the second set in the equation (28).

Case 3: T is in the rectangle formed by E and R, and has y-coordinate less than k

In this case $AE_x = AT_x - TE_x$ and $AE_y = TE_y + AT_y$, which reduces equation (34) to

$$TE_y < \frac{\text{Gain}_{\max}(A)}{2}.$$

This accounts for the third set in the (28).

Case 4: T is in the rectangle formed by E and S, and has y-coordinate less than k

In this case $AE_x = AT_x + TE_x$ and $AE_y = TE_y + AT_y$, which reduces equation (34) to

$$TE_x + TE_y < \frac{\text{Gain}_{\max}(A)}{2}.$$

This accounts for the fourth set in the (28).

Case 5: T has y-coordinate greater than or equal to k

In this case $TE_y = AE_y + AT_y$, which reduces equation (34) to

$$2AE_y - \text{Gain}_{\max}(A) < AT_x - (AE_x + TE_x) \leq 0, \tag{35}$$

where the last inequality follows from the triangle inequality. However, using equation (29), for $k \geq y_F$ we have

$$\begin{aligned} 2AE_y - \text{Gain}_{\max}(A) &= 2(k - y_E) - \text{Gain} \\ &\geq (y_F - y_E) - (x_E - x_F) + 1 \\ &= \text{Gain}' + 2 > 0, \end{aligned} \tag{36}$$

and for $k < y_F$ we have

$$\begin{aligned} 2AE_y - \text{Gain}_{\max}(A) &= 2(k - y_E) - \text{Gain} + 2(y_F - k) \\ &= (y_F - y_E) - (x_E - x_F) + 1 \\ &= \text{Gain}' + 2 > 0, \end{aligned} \tag{37}$$

which contradicts equation (35). Therefore Case 5 is impossible.

Since the five cases cover the entire node set, the necessary and sufficient conditions for $T \in R_A$ are characterized, and this completes the proof. □

5.4 Technical lemmas for the proof of Theorem 3

Lemma 3. Under Assumptions 1 and 2, the metric dimension of G' is at least 3.

Proof. Suppose that there exists two points X and Y that distinguish all points in the grid. By Claim 12, they have to be on opposite boundaries. Next, their maximum gains cannot exceed 1. Indeed, suppose for contradiction that $\text{Gain}_{\max}(X) > 1$, and that $X \in R_F$. Then, by Remark 4 we have $XF - (1 + XE) > 1$, and thus the four neighboring vertices of F will all have have distance $\min(XF \pm 1, XE + 2) = XE + 2$ to X. By Remark 6, the four neighboring vertices of F cannot be distinguished by a single vertex Y, which contradicts our assumption that $\{X, Y\}$ is a resolving set, and hence we must have $\text{Gain}_{\max}(X) \leq 1$. By a symmetric argument, we also have $\text{Gain}_{\max}(Y) \leq 1$

We have two cases depending on the parity of Gain$'$.

Case 1: Gain$'$ is even.

By Claim 13, we know that $\text{Gain}_{\max}(X)$ is also even, and since $\text{Gain}_{\max}(X) \leq 1$, it must equal to 0, which in turn implies that X is a normal vertex. By a symmetric argument, Y must be normal vertex too. Moreover, recall that X and Y must lie on opposite boundaries. Therefore, because of Claim 14, X is either $(1, \alpha - 1)$ or $(1, \alpha)$ and Y is either $(n, \beta - 1)$ or (n, β), as these are the only normal vertices on the boundaries of G'. As X and Y are normal vertices, edge e has no effect on the distances from any vertex of G' to X and Y, which therefore remain the same as in the original grid G. But we know that the only resolving sets of the grid G that have cardinality 2 are two adjacent corners of G, which disqualifies X and Y from being a resolving set of G and thus G'.

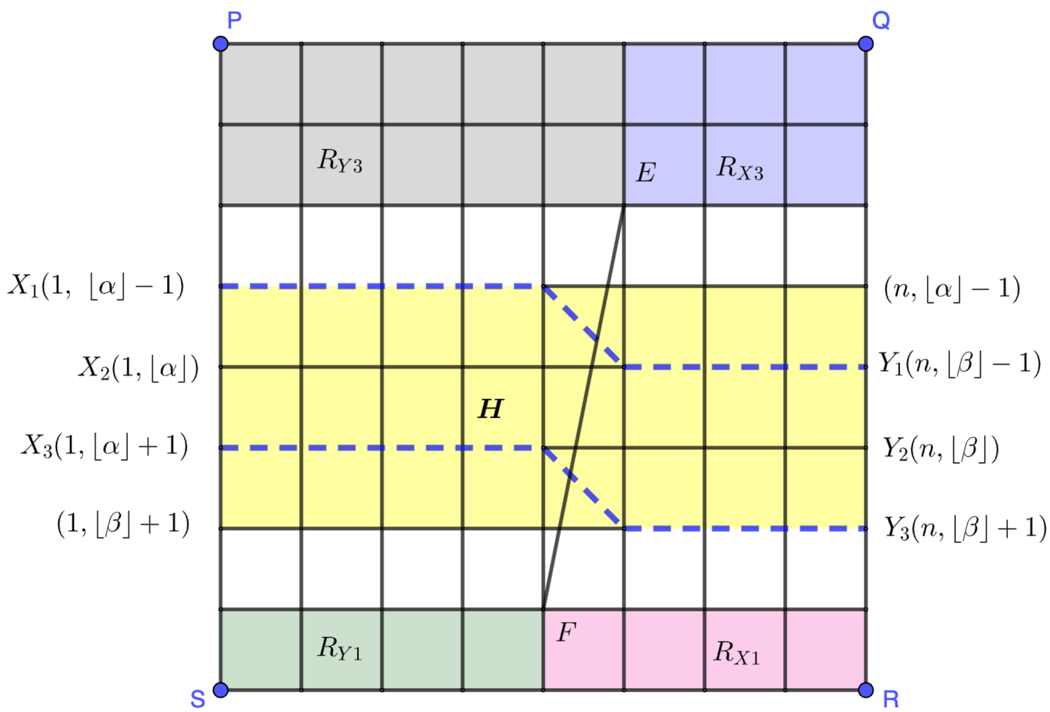


Figure 9: Figure for the proof of Lemma 3 for the case when Gain$'$ is odd. Note that sub-grid H doesn't intersect with any R_{Y_i} or any R_{X_i} for $i = 1, 3$.

Case 2: Gain$'$ is odd.

Recall, that $\text{Gain}_{\max}(X) \leq 1$, $\text{Gain}_{\max}(Y) \leq 1$, and both X and Y must lie on the boundary of G'. Let us first rule out the possibility of X or Y being on the top/bottom boundaries PQ and RS. More specifically, we will show that there is no point $X = (k, 1)$ with $\text{Gain}_{\max}(X) \leq 1$. If $X = (k, 1)$, then $X \in R_F$ and

$$\begin{aligned}
\text{Gain}_{\max}(X) &= XF - XE - 1 \\
&= |k - x_F| + y_F - 1 - |k - x_E| - (y_E - 1) \\
&= (|k - x_F| - |k - x_E|) + y_F - y_E - 1 \\
&\geq -(x_E - x_F) + y_F - y_E - 1 \\
&= \text{Gain}',
\end{aligned} \tag{38}$$

where the inequality follows from the triangle inequality. Now, Assumption 2 states that Gain$' \geq 2$, and thus no point $X = (k, 1)$ can have $\text{Gain}_{\max}(X) \leq 1$.

Now we consider the case when X and Y lie on PS and QR, respectively. We will check which vertices $X = (1, k)$ and $Y = (n, k)$ have $\text{Gain}_{\max} \leq 1$. The $\text{Gain}_{\max}$ of vertices $X = (1, k)$ is expressed in equation (29). Since Gain $>$ Gain$' \geq 2$, only the Gain $- 2(y_F - k)$ term can equal 1. The term Gain $- 2(y_F - k)$ is an increasing linear function of k that takes the value 1 for only a single value of k, namely $k = \alpha + 1/2 = \lfloor\alpha\rfloor + 1$. Similarly, in (31), the only value that satisfies $\text{Gain}' - 2(k - y_E) = 1$ is $k = \lfloor\alpha\rfloor - 1$. Consequently, the only vertices on PS that have $\text{Gain}_{\max}(X) = 1$ are $X_1 = (1, \lfloor\alpha\rfloor - 1)$ and $X_3 = (1, \lfloor\alpha\rfloor + 1)$. Similarly, the only vertices on QR that have $\text{Gain}_{\max}(X) = 1$ are $Y_1 = (n, \lfloor\beta\rfloor - 1)$ and $Y_3 = (n, \lfloor\beta\rfloor + 1)$. The only vertices on PS and QR that have $\text{Gain}_{\max}(X) = 0$ are the normal vertices $X_2 = (1, \lfloor\alpha\rfloor)$ and $Y_2 = (n, \lfloor\beta\rfloor)$. Hence, we have

$$X \in \{X_1 = (1, \lfloor\alpha\rfloor - 1), X_2 = (1, \lfloor\alpha\rfloor), X_3 = (1, \lfloor\alpha\rfloor + 1)\},$$

and,

$$Y \in \{Y_1 = (n, \lfloor\beta\rfloor - 1), Y_2 = (n, \lfloor\beta\rfloor), Y_3 = (n, \lfloor\beta\rfloor + 1)\}.$$

To finish the proof, we are going to rule out the remaining nine resolving sets that can be formed by X_1, X_2, X_3 and Y_1, Y_2, Y_3. Since $\text{Gain}_{\max}(X_1) = \text{Gain}_{\max}(X_3) = 1$, the expression for the special regions of X_1 and X_3 in Claim 15 simplifies to

$$R_{X_1} = \{(x, y) \mid x_F \leq x, y_F \leq y\} \tag{39}$$

and

$$R_{X_3} = \{(x, y) \mid x_E \leq x, y \leq y_E\}. \tag{40}$$

By a symmetric argument,

$$R_{Y_1} = \{(x, y) \mid x \leq x_F, y_F \leq y\} \tag{41}$$

and

$$R_{Y_3} = \{(x, y) \mid x \leq x_E, y \leq y_E\}. \tag{42}$$

Consider the rectangular sub-grid H formed by points X_1, $(n, \lfloor\alpha\rfloor - 1)$, Y_3, $(1, \lfloor\beta\rfloor + 1)$ (see Figure 9). The sub-grid H cannot intersect the special regions of X_1, X_3, Y_1 and Y_3 because (i) by equations (39) and (41), the special regions of X_1 and Y_1 have y-coordinate at least y_F, (ii) by equations (40) and (42), the special regions of X_3 and Y_3 have y-coordinate at most y_E, and (iii) the sub-grid H has y-coordinates more than y_E and less than y_F. The statement (iii) follows by equation (19) and the inequality $\text{Gain}' \geq 2$ from Assumption 2, since the lowest y-coordinate value of a vertex in H is

$$\lfloor\alpha\rfloor - 1 = \left\lfloor y_E + \frac{\text{Gain}' + 2}{2} \right\rfloor - 1 \geq y_E + \frac{\text{Gain}' + 1}{2} - 1 > y_E,$$

and by equation (20) and the inequality $\text{Gain}' \geq 3$ (which follows from the assumption that Gain' is odd in addition to Assumption 2), since the highest y-coordinate value of a vertex in H is

$$\lfloor\beta\rfloor + 1 = \left\lfloor y_F - \frac{\text{Gain}'}{2} \right\rfloor - 1 \leq y_F - \frac{\text{Gain}'}{2} - 1 < y_F. \tag{43}$$

Consequently, the distances in graph G' between any point in H and X or Y are same as in G. By Remark 7, we also have $\lfloor\alpha\rfloor < \lfloor\beta\rfloor$, which implies that X and Y cannot be adjacent corners of H, and they cannot resolve the sub-grid H.

Since we ruled out every pair of vertices X, Y for being a resolving set, the proof is concluded. □

Lemma 4. Under Assumptions 1 and 2, if Gain' is odd, the set $\{X = (1, \lfloor\beta\rfloor), Y = (n, \lfloor\beta\rfloor), Q = (n, 1)\}$ is a resolving set in G'.

Proof. By Claim 14, Y is a normal vertex. The only normal vertex on boundary PS is vertex $(1, \lfloor\alpha\rfloor)$, and since by Remark 7 we have $\lfloor\beta\rfloor > \lfloor\alpha\rfloor$, X cannot be a normal vertex. By Claim 8 we have $X \in R_E$, and by Remark 3, vertex X has non-empty special region $R_X \subseteq R_F$ (see the pink region in Figure 10).

Suppose for contradiction that there exist two distinct points A and B, which are not distinguished by the three points X, Y, Q in G'. We separate three cases depending on the position of A and B:

Case 1: One of A and B is in R_X, and the other is in N_X

Without loss of generality, we assume $A \in R_X$ and $B \in N_X$.

Since Y is a normal point and it does not distinguish $A = (x_A, y_A)$ and $B = (x_B, y_B)$, we have $AY = BY$, which can be expanded as

$$n - x_A + |y_A - \lfloor\beta\rfloor| = n - x_B + |y_B - \lfloor\beta\rfloor|,$$

whence

$$|y_A - \lfloor\beta\rfloor| - |y_B - \lfloor\beta\rfloor| = x_A - x_B. \tag{44}$$

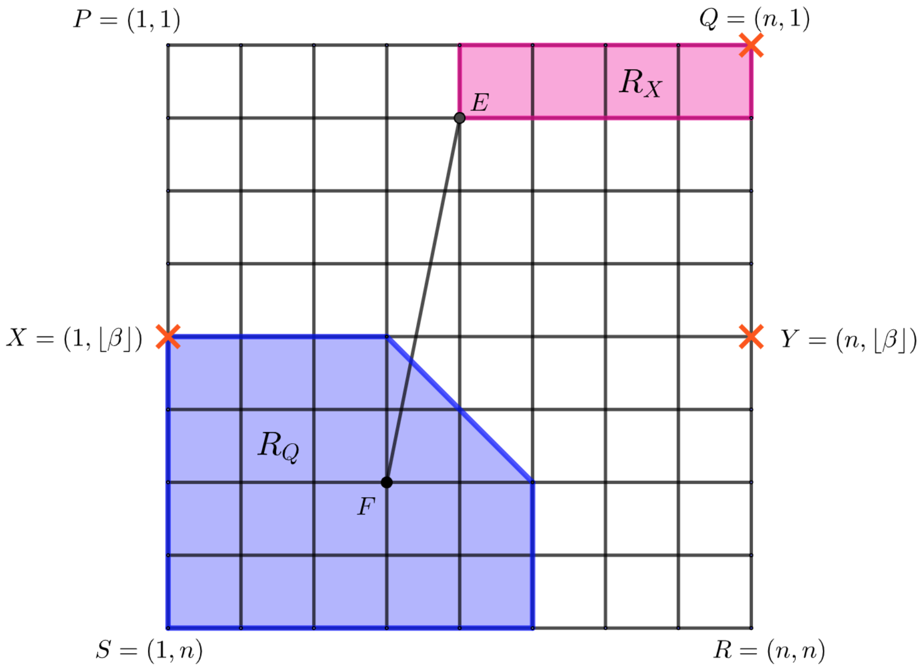


Figure 10: This is the illustration for the proof of Lemma 4. Points X, Y, Q marked with red cross form a resolving set. Y is a normal point. Blue and pink regions(boundaries included) are special regions of Q and X, respectively.

By the assumption that A and B are not distinguished by X, we have that $d_{G'}(A, X) = d_{G'}(B, X)$. Since $A \in R_X$ and $B \in N_X$, this yields that

$$AX - \text{Gain}(A, X) = BX. \tag{45}$$

Therefore,

$$\begin{aligned} \text{Gain}(A, X) &= AX - BX \\ &= (x_A - 1) + |y_A - \lfloor\beta\rfloor| - (x_B - 1) - |y_B - \lfloor\beta\rfloor| \\ &= 2(x_A - x_B), \end{aligned} \tag{46}$$

where the last line follows form equation (44). Equation (46) implies that $\text{Gain}(A, X)$ must be even. By Claim 13, if $\text{Gain}(A, X)$ is even then Gain' must be even too, which contradicts our assumption that Gain' is odd.

Case 2: $A, B \in N_X$

In this case, the distances between X, Y and A, B are the same in graph G' as in G, which implies that A, B are not distinguished by X nor Y in G. The only pairs of vertices that are not distinguished by X, Y in the grid G are vertices that are symmetric to the horizontal line passing through X and Y. Therefore A, B must be such a pair. By a similar parity based argument as in Case 1, if one of A and B is in R_Q and the other is not, then they are distinguished by either Y or Q. Indeed, substituting Q instead of X into equations (45) and (44), we get

$$\begin{aligned} \text{Gain}(A, Q) &\overset{(45)}{=} AQ - BQ \\ &= n - x_A + y_A - 1 - (n - x_B) - (y_B - 1) \\ &\overset{(44)}{=} |y_B - \lfloor\beta\rfloor| - |y_A - \lfloor\beta\rfloor| + y_A - y_B \\ &\equiv 0 \pmod 2. \end{aligned} \tag{47}$$

Then, Gain' should also be even by Claim 13, contradicting our assumption that Gain' is odd.

We are left with the cases $A, B \in N_Q$ and $A, B \in R_Q$. Notice that since we showed $\lfloor\beta\rfloor + 1 < y_F$ for odd Gain' in equation (43), and since by equation (20) we have $\lfloor\beta\rfloor = \lfloor y_E + (\text{Gain} + 2)/2\rfloor > 1$, neither F nor Q are on the horizontal line through X and Y. Hence, any pair of nodes A, B that are symmetric to the XY line are distinguished by both Q and F in graph G. We immediately see that if $A, B \in N_Q$, the pair A, B is also by Q in G'. If $A, B \in R_Q$, by Claim 3 together with $Q \in R_F$, and since F distinguishes A, B in G, we have

$$d_{G'}(Q, A) = QE + 1 + FA \neq QE + 1 + FB = d_{G'}(Q, B).$$

Hence, in every sub-case of Case 2 we showed that A, B must be distinguished by at least one of Q, X and Y in G'.

Case 3: $A, B \in R_X$:

By Claim 15, we have $Q \in R_X$. The anti-transitivity property of special regions (Remark 2) implies that if $A \in R_X$ and $Q \in R_X$, then $A \notin R_Q$ and therefore $A \in N_Q$. Similarly, we have $B \in N_Q$, and we can deduce that the distances between Q and A, B are the same in graph G' as in graph G. Moreover, since Y is a normal vertex, the distances distances between Y and A, B are the same in G' as in G too.

Remark 3 together with $X \in R_E$ implies that we have $R_X \subseteq R_F$, and Remark 8 and Claim 14 together imply that every vertex in R_F has y-coordinate at most $\beta - 1 < \lfloor \beta \rfloor$. Hence, both A and B are contained in the rectangular sub-grid with corners $QYXP$. Since Q and Y are adjacent corners of the sub-grid $QYXP$, they must resolve the entire sub-grid $QYXP$ in graph G, including vertices A and B. Since distances from Y and Q to A, B are the same in graph G' as in G, vertices Q and Y must distinguish A and B in G' as well.

Thus, every vertex pair A, B is distinguished by some vertex in the set $\{X, Y, Q\}$, and the proof is concluded. $\square$

Lemma 5. Under Assumptions 1 and 2, if Gain$'$ is even and $x_E - x_F < \frac{\text{Gain}'}{2} + 2$, the set $\{X = (1, \beta - 1), Y = (n, \beta - 1), Z = (1, \alpha - 1)\}$ is a resolving set in G'.

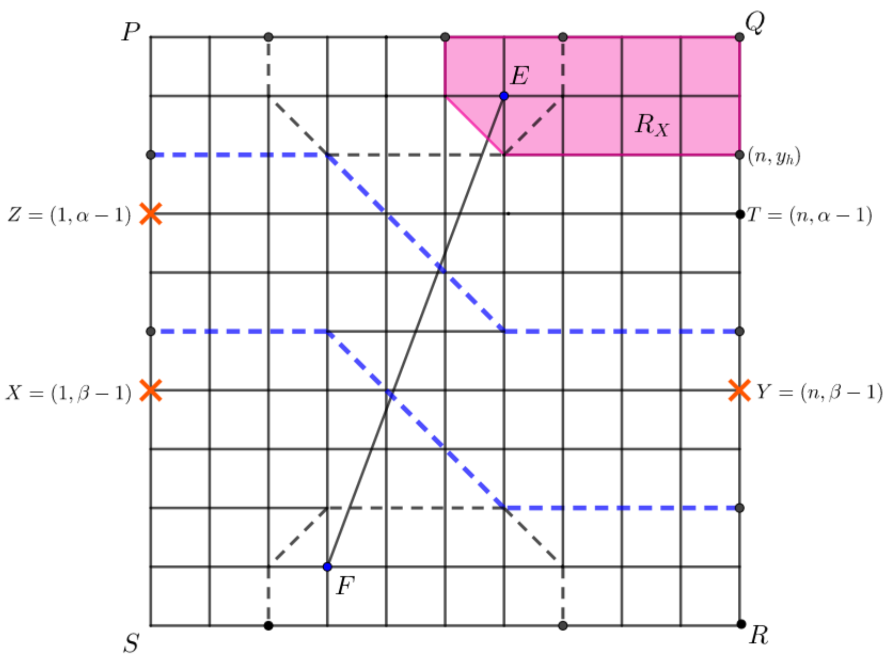


Figure 11: Illustration for the proof of Lemma 5. Points X, Y, Z marked with red crosses form a resolving set. Y and Z are normal points, the pink region(including the boundary) is R_X.

Proof of Lemma 5. See Figure 11 for an illustration. Note that Y and Z are normal points, and that $X \in R_E$. First we calculate $\text{Gain}_{\max}(X)$. By equation (29), since $\beta - 1 \leq y_F$ because of equation (20),

$$\begin{aligned} \text{Gain}_{\max}(X) &= \text{Gain} - 2(y_F - \beta + 1) \\ &= 2(x_E - x_F) - 2. \end{aligned} \tag{48}$$

Now we show that R_X completely lies inside the rectangle $PQTZ$. Indeed, according to Claim 15, the largest y-coordinate of a point in special region of X will be

$$
\begin{aligned}
y_{\max} &= y_E + \frac{\mathrm{Gain}_{\max}(X)}{2} - 1 \\
&\overset{(48)}{=} y_E + \frac{2(x_E - x_F) - 2}{2} - 1 \\
&= \frac{1 + y_F + y_E + x_F - x_E}{2} - 1 - \frac{(y_F - y_E) - (x_E - x_F) - 1}{2} + (x_E - x_F) - 2 \\
&= \alpha - 1 - \frac{\mathrm{Gain}'}{2} + (x_E - x_F) - 2 \\
&< \alpha - 1, \qquad (49)
\end{aligned}
$$

where the inequality follows by the assumption $x_E - x_F < \mathrm{Gain}'/2 + 2$. Hence, since we also have $\alpha < \beta$ by Remark 7, all points in the special region of X will have y-coordinate less than that of Z. Alternatively, denoting vertex $(n, \alpha - 1)$ by T, we have that R_X is contained in the rectangle $PQTZ$.

Let us suppose for contradiction that there exist two distinct points $A = (x_A, y_A)$ and $B = (x_B, y_B)$ which are not distinguished by X, Y, Z. We distinguish three cases based on the positions of A and B:

Case 1: $A, B \in N_X$

In this case all distances between X, Y, Z and A, B are the same in graph G' as in graph G. It is easy to see that to be equidistant from X and Y, vertices A and B must be symmetric to the horizontal line through X and Y, in which case Z can distinguish A and B.

Case 2: $A, B \in R_X$

In this case, we show that A and B cannot be equidistant from both Y and Z. Both A and B lie inside of R_X, and thus the region $PQTZ$. Now we show that Y and Z resolve $PQTZ$ in G, which implies that they resolve $PQYZ$ in G' because they are normal vertices. Our argument will be similar to the standard argument that shows that two adjacent corners resolve the grid. To be equidistant from Z, both of them should lie on a diagonal line parallel to PR, or equivalently,

$$x_A - y_A = x_B - y_B. \qquad (50)$$

To be equidistant from Y, they should lie on a diagonal line parallel to QS, or equivalently,

$$x_A + y_A = x_B + y_B. \qquad (51)$$

However, equations (50) and (51) cannot hold simultaneously for $A \neq B$.

Case 3: One of A and B is in R_X, and the other is in N_X

Without loss of generality, we assume that $A \in R_X$ and $B \in N_X$. Since R_X lies inside of $PQTZ$, we know that the y-coordinate of A is less than that of Z and Y, i.e., $y_A < \alpha - 1 < \beta - 1$. Since we have shown in Case 2 that Y and Z resolve $PQTZ$ in G', B cannot lie in the region $PQTZ$. There are two other possibilities for where B could lie:

1. Let us assume that B lies in the region $ZTYX$. Since Y does not distinguish A and B, we have $AY = BY$, which implies that $x_A + y_A = x_B + y_B$ as in equation (51). Similarly, since Z does not distinguish A and B, we have $AZ = BZ$, which implies that $x_A + (\alpha - 1 - y_A) = x_B + y_B - (\alpha - 1)$. Subtracting the second equation from the first gives $y_A = \alpha - 1$ which contradicts equation (49).

2. Let us assume that B lies in the region $XYRS$, or equivalently, $y_B \geq \beta - 1$. Since we have assumed A and B to be equidistant from X and Z, we have $AZ = BZ$ and $BX = d_{G'}(A, X) = AX - \mathrm{Gain}(A, X)$. Writing these equations in terms of the variables x_A, y_A, x_B and y_B gives

$$x_A - 1 + (\alpha - 1 - y_A) = x_B - 1 + y_B - (\alpha - 1), \qquad (52)$$

and

$$x_A - 1 + (\beta - 1 - y_A) - \mathrm{Gain}(A, X) = x_B - 1 + y_B - (\beta - 1). \qquad (53)$$

Subtracting equation (53) from equation (52) yields

$$\mathrm{Gain}(A, X) = 2(\beta - \alpha) = 2(x_E - x_F). \qquad (54)$$

Equations (54) and (48) together contradict the fact that $\mathrm{Gain}(A, X) \leq \mathrm{Gain}_{\max}(X)$.

We considered all cases and the proof is concluded. □

5.5 Precise conjecture

Finally, we present our precise conjecture which completely characterizes metric dimension for any grid graph augmented with one edge. We believe this can be proved by rigorous case-wise analysis but it is out of the scope of this paper. We have verified this conjecture for square grids with sizes up to 15×15 using simple C++ programs available at [36]. Note that the conjecture is stated not only for square grids but also for $m \times n$ rectangular grids, but for these graphs we only verified the conjecture for a few parameter values due to the increased number of cases.

Conjecture 1. Let G be a grid graph with m rows and n columns. Let e be the edge between vertices $F = (x_F, y_F)$ and $E = (x_E, y_E)$ with $x_F, x_E \in \{1, ..., n\}$, $y_F, y_E \in \{1, ..., m\}$, with the assumption that $EF \geq 2$. Let $G' = (V, E_G \cup \{e\})$ be the grid augmented with one edge. Let $\mathrm{Gain} = |y_E - y_F| + |x_F - x_E| - 1$ and $\mathrm{Gain}' = ||y_F - y_E| - |x_F - x_E|| - 1$.

- $\beta(G') = 4$ if all of the following conditions are satisfied:
  - None of the endpoints of e is a corner of the grid. i.e.,
  $$(x_E, y_E), (x_F, y_F) \notin \{(1,1), (n,1), (1,m), (n,m)\}$$
  - Gain' is positive and even.
  - $\min(|x_F - x_E|, |y_F - y_E|) \geq \frac{\mathrm{Gain}'}{2} + 2$
- $\beta(G') = 2$ if any of the following conditions is satisfied:
  - $\mathrm{Gain} = 1$
  - $\mathrm{Gain}' \leq 1$, Gain is odd and one of the endpoints is a corner of the grid.
  - $\mathrm{Gain}' \geq 3$, Gain is odd, $\mathrm{Gain} - \mathrm{Gain}' \leq 2$ and one of the endpoints is a corner of the grid.
  - Gain is odd and both endpoints are corners of the grid.
- $\beta(G') = 3$ for all other cases.

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